

Sr	Supply chain problem	Application domain	Title	Year	Venue	Specific aspect / SC aspect	Tools used to build the model	Simulation method used	Optimization/ prediction method used	Comments	Any attributes
1	Demand Forecasting; Inventory Analysis and Optimization	Other/Not mentioned	Using simulation-based system dynamics and genetic algorithms to reduce the cash flow bullwhip in the supply chain	2020	International Journal of Production Research (IJPR)	Demand forecasting, Inventory, replenishment policies	MATLAB	SD	Simulation-based optimization, SBO	Cash Flow Bullwhip (CFB) and the traditional inventory bullwhip effect (BWE), i.e., variability amplification in orders and in cash conversion cycles across stages. The paper aims to minimize CFB, BWE, and supply chain total cost (SCTC) by finding optimal control parameters.	Modeling parameters desired inventory desired supply line sales price per unit) unit cost upstream sales price demand smoothing parameter orders placed order quantity inventory level backlog financial policy (pay policy, collection policy) Performance metrics (output) Supply Chain Total Cost Bullwhip Effect variance(orders)/variance(demand) Cash Conversion Cycle (CCC) (derived metric) Cash Flow Bullwhip variance(CCC)/variance(demand) Inventory levels and backlog levels Order quantity trajectories and order variance
2	Demand Forecasting; Inventory Analysis and Optimization	Other/Not mentioned	An Agent-based Simulation Model to Mitigate the Bullwhip Effect via Information Sharing and Risk Pooling	2022	Winter Simulation Conference (WSC)	Analysis: Bullwhip effect (ripple of rising demand towards supplier in chain) Authors model 3 echelon SC to study impact of info sharing and risk pooling	AnyLogic	ABS	Analysis: simulation-based	The paper investigates strategies to mitigate the bullwhip effect in supply chains through an agent-based simulation model. It focuses on the impact of information sharing and risk pooling on reducing demand distortion across a three-echelon supply chain comprising a manufacturer, wholesaler, and two retailers	SC structure (2 retailers, a wholesaler, manufacturer) demand and order quantity inventory replenishment policy (order-up-to level, safety stock, moving average forecast) Lead time
3	Demand Forecasting; Supply Chain Disruptions	Other/Not mentioned	State of the art, conceptual framework and simulation analysis of the ripple effect on supply chains.	2022	International Journal of Production Research (IJPR)	Supply chain disruptions	Python	ABS	Analysis	The paper tackles the fragmented state of knowledge on the ripple effect: how disruptions spread through supply chains, which mitigation and recovery measures exist, and where empirical and methodological gaps remain. They performed a systematic literature review (50 selected documents) to classify research streams, methods, application areas and mitigation strategies. From the review they proposed a conceptual framework that links disruption drivers, resilience pillars (redundancy, robustness, flexibility), digital technologies, and proactive/reactive measures. They implemented and validated that framework with a System Dynamics simulation model (adapted from Gu and Gao) to quantify ripple propagation and to test how inventory cover time, recovery speed and mitigation policies affect service levels and profits.	Disruption attributes Operational parameters: lead times, processing times, transport capacity, safety stock/cover time, reorder policies. Service level; share of demand met; time-to-recover; time-to-survive; resilience indices (recovery speed and magnitude). Economic metrics: total cost, disruption cost, profit Topological/structural metrics: node connectivity, largest functional component, path lengths. Variability measures: order/inventory variance, bullwhip/ripple quantification
4	Demand Forecasting; Inventory Analysis and Optimization	Other/Not mentioned	The importance of forecast uncertainty in understanding the Bullwhip effect	2025	International Journal of Production Research (IJPR)	Analysis: how forecast uncertainty propagates upstream, measured with a proposed Ratio of Forecast Uncertainty against the bullwhip ratio and net stock amplification	R	Monte Carlo	Simulation-based analysis	An order-up-to inventory simulation with several demand processes and forecasting models, 500 replications per process. The new metric relates more strongly to upstream inventory cost than standard bullwhip	demand process, forecasting model, lead time, order-up-to policy; performance: bullwhip ratio, net stock amplification, ratio of forecast uncertainty, inventory
5	Energy and Carbon Cost Modeling and Optimization	Industrial and Manufacturing	How will CBAM affect the decarbonisation of steel industry in China? A system dynamics approach	2024	International Journal of Production Research (IJPR)	Analysis: impact of the EU Carbon Border Adjustment Mechanism on steel exports, profit and carbon emissions, and of decarbonisation policies	Not specified	SD	Simulation-based scenario analysis	Builds a causal loop and stock-flow model of the Chinese steel SC, introduces CBAM as an exogenous shock and evaluates decarbonisation scenarios. CBAM cuts export volume by 32% and export profit by 58% with limited environmental benefit.	material flows, export volume, carbon price, emission intensity, decarbonisation measures; performance: export profit, export volume, carbon emissions
6	Energy and Carbon Cost Modeling and Optimization	Healthcare and Medical	Towards Using Simulation to Evaluate the Circular Economy of Small Medical Devices	2024	Winter Simulation Conference (WSC)	Analysis: environmental, operational and financial effect of circular-economy interventions (reuse, refitting) for laparoscopic scissors	AnyLogic	DES	Simulation-based scenario comparison	Compares linear and circular flows from production to end of life. Preliminary results: circular flows reduce emissions overall and improve cost-effectiveness from the hospital's perspective.	production, storage, dispatch to hospital, use cycles, recovery and refitting rates, emissions factors, costs; performance: emissions, cost, instrument availability

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7	Energy and Carbon Cost Modeling and Optimization	Industrial and Manufacturing	A simulation-based decision support tool for circular manufacturing systems in the automotive industry using electric machines as a remanufacturing case study	2025	International Journal of Production Research (IJPR)	Analysis and optimization: economic and environmental effect of end-of-life design strategies (reuse, remanufacturing) in a circular SC for electric machines	AnyLogic	Hybrid DES + ABS	OptQuest	Parts are agents carrying material, cost and mass; manufacturing, utilisation and end-of-life handling are modelled as processes. Scania electric machine case: 10.6% more design investment cuts average cost by 18.6%, material demand by 14.7% and cradle-to-gate impact by 38.7%.	part material, cost, mass, lifetime length, degradation speed, success rate of remanufacturing and reuse, end-of-life strategy, workers; performance: total operational cost, carbon footprint, material demand
8	Inventory Analysis and Optimization; Logistics Optimization and Route Planning	Other/Not mentioned	Agri-food supply chains with stochastic demands: A multi-period inventory routing problem with perishable products	2019	Simulation Modelling Practice and Theory (SIMPAT)	Inventory and routing	Not specified	Monte Carlo	Optimization of inventory and routing	IRP, optimization of perishable inventory and routing decisions to minimize inventory, costs, food waste	inventory costs, routing costs, perishable product (and parameters)
9	Inventory Analysis and Optimization; Logistics Optimization and Route Planning	Other/Not mentioned	An Inventory-Routing Problem with Stochastic Demand and Stock-out: A Solution and Risk Analysis using Simheuristics	2019	Winter Simulation Conference (WSC)	Inventory routing problem (IRP) is addressed (choosing optimum route to maintain inv levels at diff nodes when transporting goods to minimize inventory and routing costs). choose optimum route to minimize total cost = routing cost + inventory cost	Not specified	Monte Carlo	Sim with different inv cap and replenishment level to find optimum and multi-start optimization method for IRP	Logistics and Inventory optimization Authors address inventory routing problem: optimize route and inventory replenishment levels to minimize inventory cost as well as routing cost	Yes. Inventory replenishment level, routing cost, inv hold cost, inv stockout cost
10	Inventory Analysis and Optimization; Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	Inventory Management under Disruption Risk	2019	Winter Simulation Conference (WSC)	Analysis: Inventory stocking decisions in presence of disruption	SAS Simulation Studio	Hybrid DES + Monte Carlo	SAS software's ML module is used to do analysis of simulation data, and report results.	Authors combine newsvendor model and disruption model to do analysis of inventory stocking decisions when SC is subjected to disruption (natural disaster/man made disruption). They discuss parameters like disruption impact on inventory, disruption probability, inv loss due to disruption and so on.	Yes. Demand, p: prob of disruptive events l: inv loss due to disruptive event inv hold cost, inv shortage cost, inv loss cost (due to disruption), inv recovery cost (to recover from disruption loss), inv capacity
11	Inventory Analysis and Optimization	Other/Not mentioned	Simulating Profit Loss in Behavioral Newsvendor Problems	2019	Winter Simulation Conference (WSC)	optimum inventory levels in newsvendor problem that maximizes the profit	Not specified	Monte Carlo	Different scenarios are simulated to do sensitivity analysis	Authors use data from newsvendor experiments to calculate parameters for a model to predict the average ordering behavior of suboptimal decision makers. This model is used for simulation: examine how changes in the profit margin, price, and demand changes affect the financial risk	Yes. cost, profit, inventory Cost,
12	Inventory Analysis and Optimization	Other/Not mentioned	Simulation of Allocation Policies for a Serial Inventory System under Advance Demand Information	2019	Winter Simulation Conference (WSC)	inventory allocation policy selection to minimize inventory cost	AnyLogic	DES	Different allocation policies are simulated and performance is assessed	ADI advance demand info is available, Order base-stock policy is used: the inventory is replenished after the order is placed by a customer. (Making reservations/reserving your product, and then it is shipped to you) Allocation policies: First come first serve, Priority, Inventory Rationing, Last-minute ...	Yes. Poisson demand. Lead time. Inventory parameters: back order, level
13	Inventory Analysis and Optimization	Other/Not mentioned	A Deep Reinforcement Learning Approach For Optimal Replenishment Policy In A Vendor Managed Inventory Setting For Semiconductors	2020	Winter Simulation Conference (WSC)	Inventory optimization (choosing optimal replenishment policy)	AnyLogic	ABS	Reinforcement Learning	Authors propose to model and simulate Vendor Managed Inventory (VMI) approach, and use reinforcement learning to select optimum inventory replenishment policy to maximize the performance of the system	Yes. Not explicitly mentioned
14	Inventory Analysis and Optimization; Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	Inventory Management with Disruption Risk	2020	Winter Simulation Conference (WSC)	Analysis: Inventory stocking decisions in presence of disruption	SAS Simulation Studio	Hybrid DES + Monte Carlo	SAS software's ML module is used to do analysis of simulation data, and report results.	Authors combine newsvendor model and disruption model to do analysis of inventory stocking decisions when SC is subjected to disruption (natural disaster/man made disruption). They discuss parameters like disruption impact on inventory, disruption probability, inv loss due to disruption and so on.	Yes. Demand, prob of disruptive events inv loss due to disruptive event inv hold cost, inv shortage cost, inv loss cost (due to disruption), inv recovery cost (to recover from disruption loss), inv capacity, base stock level
15	Inventory Analysis and Optimization	Healthcare and Medical	Scenario-based Simulation Approach for an Integrated Inventory Blood Supply Chain System	2020	Winter Simulation Conference (WSC)	Simulation model to support management (or to support critical decisions in the system)	Arena	DES	Simulation-based	Authors simulate Blood supply chains (network of blood banks, blood collection centers, hospitals and logistics to transport blood from place to place) using DES in Arena. They support modeling of blood shelf life, cross matching, donation methods... They propose integrated inventory systems (IIS) (including vehicle depot to facilitate the flow of blood products) that connects blood banks and hospitals	Inventory policy: FIFO, specific parameters: blood donation method, Blood product type (RBC, plasma, whole, platelets)...

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16	Inventory Analysis and Optimization	Other/Not mentioned	A dynamic model for deteriorating products in a closed-loop supply chain	2021	Simulation Modelling Practice and Theory (SIMPAT)	Analysis	Vensim	SD	Simulation-based analysis	They present a dynamic model for managing deteriorating products in a closed-loop supply chain involving a manufacturer and a distribution center. The study simulates the impact of various strategies (such as bargaining, better warehousing, and changing collection rules) on key performance indicators like total cost, total sales, and total disposal. The results indicate that combining better warehousing conditions with more frequent collections yields the best performance. The study highlights the importance of inventory management for deteriorating products and provides insights into optimizing supply chain operations through simulation	production rate, deterioration rate, shipment rate, demand, inventory levels total sc cost, sale, disposal
17	Inventory Analysis and Optimization; Resilience Modeling, Risk Estimation and Mitigation	Healthcare and Medical	Improving Simulation Optimization Run Time When Solving for Periodic Review Inventory Policies in a Pharmacy	2021	Winter Simulation Conference (WSC)	inventory optimization (s,S) to minimize per day costs	Not specified	Other	Simulation-based	simulation model accounts for perishability, positive lead time, stochastic demand, and supply disruptions. They propose Binary-Grid-Search algorithm to quickly search for optimum.	Inventory parameters (s,S), lead time, shelf life, inventory shortage, waste, holding, ordering cost, purchase cost
18	Inventory Analysis and Optimization	Healthcare and Medical	Measuring the Impact of Data Standards in an Internal Hospital Supply System	2021	Winter Simulation Conference (WSC)	Analysis: simulation to illustrate the potential benefits of using data standards within a hospital supply system in terms of the improved management of items that can expire.	Arena	DES	Simulation-based analysis	In hospital SC, using data standards proves to be useful (unique identifiers for resources and inventory). Authors illustrates its benefits by simulating hospital SC with data standards. They model inventory with SKU, and shelf life for inventory items	Inventory replenishment policies: (r,Q) reorder point, reorder quantity: when reorder point is reached, place order for Q quantity (T,S) periodic review period (T), Order up to level (S): review inventory periodically, and then order to fill up to level S other parameters: Inventory with shelf life, lead time, SKU
19	Inventory Analysis and Optimization	Healthcare and Medical	BloodChainSim – a Simulation Environment to Evaluate Digital Innovations in Blood Supply Chains	2022	Winter Simulation Conference (WSC)	Optimization of inventory (and logistics) to maximize availability (to minimize blood shortage)	AnyLogic	Hybrid DES + ABS	Simulation-based	BloodChainSim is a toolkit developed on top of AnyLogic.	Not explicitly
20	Inventory Analysis and Optimization	Industrial and Manufacturing	Dynamic inventory replenishment strategy for aerospace manufacturing supply chain: combining reinforcement learning and multi-agent simulation	2022	International Journal of Production Research (IJPR)	Inventory (Production and assembly processes, Transportation and shipment batching, Multi-echelon material and part flows)	AnyLogic	ABS	OptQuest	replace static (I, R, S) periodic review parameters with a dynamic policy learned via reinforcement learning to maximize overall supply chain performance (SCP) in the presence of stochastic demand, variable lead times, and complex production processes.	Parameters (input): inventory levels, processing times demand: interarrival distributions, order quantities Transportation: shipment volumes, transportation times Cost parameters: unit manufacturing cost, raw material cost, inventory holding cost, transportation cost, tardiness fine, product prices. Performance metrics and outputs revenue, manufacturing cost, transportation cost, inventory holding cost, tardiness fines, raw material cost. (total/avg), Optimal policy
21	Inventory Analysis and Optimization; Energy and Carbon Cost Modeling and Optimization	Other/Not mentioned	Effect of load bundling on supply chain inventory management: An evaluation with simulation-based optimisation	2022	Journal of Simulation (JoS)	Inventory	AnyLogic	DES	Simulation-based optimization with an evolutionary metaheuristic (NSGA-II); multicriteria objective trading off overall cost against service level, with CO2 emissions also evaluated	Generic multi-stage, multi-item supply chain (one manufacturer, one distribution centre, several retailers, 10 products). Evaluates the effect of load bundling at the distribution centre on overall costs, service level and CO2 emissions. Load bundling yields smaller and more customer-driven lot sizes which simultaneously reduce carbon emissions; for medium to high service level targets, ABC-clustered order rate scenarios give lower supply chain costs than an identical order rate for all items.	(s, Q) inventory policy; decision variables: reorder point s and lot size Q for each product at each supply chain member; product demand arrival rates (25, 7.5 and 1.67 pieces/day); ABC-clustered vs identical order rates; load bundling opportunity at the distribution centre; target service level performance: overall supply chain cost, service level (item availability at the retailers), CO2 emissions, lot size

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22	Inventory Analysis and Optimization	Industrial and Manufacturing	Inventory policy for postponement strategy in the semiconductor industry with a die bank	2022	Simulation Modelling Practice and Theory (SIMPAT)	Inventory optimization	Arena	DES	Simulation-based	discusses an inventory policy for the semiconductor industry, focusing on the use of a die bank (intermediate fabricated wafers known as dies) to improve inventory management and customer responsiveness. By storing intermediate forms of wafers, the die bank allows for postponement of product differentiation, reducing finished goods inventory and adapting more flexibly to demand changes.	demand, forecast, inventory level, inventory quantity, production lead time avg inv levels, service level, fill rate, cycle service time
23	Inventory Analysis and Optimization	Healthcare and Medical	On perishable inventory in healthcare: random expiration dates and age discriminated demand	2022	Journal of Simulation (JoS)	Inventory optimization	MATLAB	Monte Carlo	Simulation-based optimization (Simulated Annealing)	Perishable inventory of blood platelets. Addresses random premature expiration, random demand, irregular supply and age-differentiated (age-discriminated) demand under custom medical compatibility guidelines. Simulation-optimization with new settings and replenishment policies determines the optimized daily order quantity for this combination of inventory restrictions.	daily order quantity (decision variable); random and premature expiration dates; platelet age settings and shelf life; stochastic daily demand; age-discriminated demand; irregular supply; custom replenishment guidelines performance: objective function value evaluated by stochastic simulation within the Simulated Annealing search
24	Inventory Analysis and Optimization; Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Simulation-Based Analysis Of Recovery Actions Under Vendor-Managed Inventory Amid Black Swan Disruptions In The Semiconductor Industry: A Case Study From Infineon Technologies AG	2022	Winter Simulation Conference (WSC)	Analysis: performance of Vendor Managed Inventories (VMI) during disruptions	AnyLogic	Hybrid DES + ABS	Analysis: simulation-based	The paper presents a simulation-based analysis of recovery actions under vendor-managed inventory (VMI) amid black swan disruptions in the semiconductor industry. It explores the impact of the COVID-19 pandemic on supply chains, particularly focusing on the automotive sector semiconductor supply chain. Black Swan Disruptions refer to highly unpredictable and rare events that have severe consequences.	Demand, Lead time, Inventory parameters (capacity, costs) Inventory levels (at retailer, warehouse, manufacturer, supplier) replenishment policy (R,s,S): R review period, s reorder point, S reorder level Production capacity, Disruption phase
25	Inventory Analysis and Optimization	Industrial and Manufacturing	The effects of pre or post delay variable changes in discrete event simulation and combined DES/system dynamics approaches in modelling supply chain performance	2022	Journal of Simulation (JoS)	pre or post delay variable changes in DES	Arena	Hybrid DES + SD	No optimization. Analysis: comparison of three modeling approaches (pre-delay DES, post-delay DES, combined DES/SD)	Cement supply chain (Jordan). Compares pre-delay DES, post-delay DES and a combined DES/SD model of the same supply chain. Post-delay DES overestimates inventory level and holding cost but underestimates ordering cost; pre-delay DES does the reverse; the combined DES/SD model falls between the two.	inventory level (IL), inventory position (IP), backorders (BO), scheduled orders (SO); holding cost rate about 3% of cement price per ton per day; ordering cost; demand; delays; FCFS priority rule; backordering without penalty cost performance: inventory level, holding cost, ordering cost
26	Inventory Analysis and Optimization	Other/Not mentioned	A Deep Q-Network Based on Radial Basis Functions for Multi-Echelon Inventory Management	2023	Winter Simulation Conference (WSC)	Optimization: inventory policy	Python	Other	Deep Q Networks and RBF	proposed approach is simulated in Python to assess its performance in multi-echelon supply chain. Authors propose Deep Q net and RBF to model supply chains (multi-echelon inventory problem), to optimize replenishment policy	demand, Lead time, Inventory levels, costs (shortage, order, holding)
27	Inventory Analysis and Optimization	Agriculture and Food	A simulation-optimisation approach for production control strategies in perishable food supply chains	2023	Journal of Simulation (JoS)	Inventory optimization	Arena	DES	Simulation-based optimization (OptQuest in Arena: tabu search, neural networks, scatter search); objective: minimize total cost subject to a target service level	Food SC (three-echelon perishable food supply chain). Authors propose B-CONWIP (Basestock-Constant Work-in-Process), a pull-based inventory control policy, and compare it with continuous review (s, S) and BSP-low-EW (periodic review). Objective: minimize total cost (holding + deterioration + ordering + shortage) while meeting a 99% service level. B-CONWIP gives the lowest total cost and is the most robust to demand rate and demand volume variations.	control parameters: basestock level, reorder point s, break point b, number of CONWIP cards; demand rate and demand mean variations (5% to 25%); product deterioration / limited shelf life; target service level 99% performance: holding cost, deterioration cost, ordering cost, shortage cost, total cost (objective function), service level

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28	Inventory Analysis and Optimization; Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Applying digital twins for inventory and cash management in supply chains under physical and financial disruptions	2023	International Journal of Production Research (IJPR)	Inventory Supply chain disruptions	SimPy	DES	Analysis	Problem: The paper examines how simultaneous physical and financial disruptions propagate through a three-tier supply chain and how downstream payment policies can squeeze upstream liquidity, worsening resilience and operational performance. How it was addressed: The authors built a SimPy discrete-event digital twin of a three-echelon supply chain to run scenario experiments (payment increases, demand surges, capacity drops). They then trained a CART decision tree on simulation outputs (10,000+ generated examples) to produce interpretable decision rules that recommend inventory and cash replenishment settings to keep upstream CCC within acceptable bands. Results show payment policy shifts materially change CCC across tiers, inventory levels dominate CCC effects under physical shocks, and the decision-tree rules enable centralized planning to outperform decentralized policies in many scenarios.	Parameters Inventory policy parameters: desired inventory (DI), desired WIP inventory Financial policy parameters: credit purchase policy n and credit collection policy m Operational parameters: manufacturer capacity, customer demand, lead times Performance: Cash Conversion Cycle (CCC) per echelon Service level (retailer fill rate), inventory levels
29	Inventory Analysis and Optimization; Resilience Modeling, Risk Estimation and Mitigation	Healthcare and Medical	Effects of Timing of Agents' Reactions in Pharmaceutical Supply Chains under Disruption	2023	Winter Simulation Conference (WSC)	Analysis: analysis of impact of disruptions on pharma supply chains using multiagent simulation	Not specified	ABS	Analysis: simulation-based	Authors model pharma supply chains under disruptions (they model flow of product, inventory levels, trustworthiness, agent behavior). The study shows: communication can exacerbate disruption impact	Inventory parameters, Transportation parameters Performance Metrics: Trustworthiness measure, profit measure, backlog levels, inventory costs
30	Inventory Analysis and Optimization; Demand Forecasting	Other/Not mentioned	The Bullwhip Effect in End-to-end Supply Chains: The Impact of Reach-based Replenishment Policies with a Long Cycle Time Supplier	2023	Winter Simulation Conference (WSC)	analysis: inventory policy impact in bullwhip effect	AnyLogic	SD	Simulation-based	Authors study how different replenishment policies impact Bullwhip Effect (BWE). They model and simulate inventory replenishment policies to show their performance analysis under BWE	VMI (vendor managed inventory), Reach-based Kanban, Absolute stock based
31	Inventory Analysis and Optimization	Industrial and Manufacturing	Aggregated Simulation Modeling to Assess Product-Specific Safety Stock Targets During Market Up- and Downswing: A Case Study	2024	Winter Simulation Conference (WSC)	Analysis: product-specific safety stock targets at the die bank and their effect on on-time delivery during market up- and downswings	Not specified	Other	Simulation-based analysis	Demand is built from forecasts adjusted by sampled historical forecast errors; only bottleneck resources are modelled, on existing master data. Forecast bias depends on market phase, so the effect of safety stock differs between upswings and downswings. The simulation paradigm is not named.	safety stock target, forecast and forecast error distribution, front-end and back-end lead times, bottleneck capacity, FCFS scheduling; performance: on-time delivery, inventory
32	Inventory Analysis and Optimization; Logistics Optimization and Route Planning	Humanitarian and Emergency	Coupling simulation and machine learning for predictive analytics in supply chain management	2024	International Journal of Production Research (IJPR)	Inventory, Logistics (Inventory allocation and safety stock, Sourcing, transport, replenishment, and distribution strategies.)	AnyLogic	Hybrid DES + ABS	Pre-crisis performance assessment of humanitarian supply chains. Coping with data scarcity, timeliness, and interpretability challenges in forecasting humanitarian logistics performance.	Authors demonstrates how coupling simulation and machine learning can overcome the data scarcity, timeliness, and interpretability issues in humanitarian supply chain predictive analytics, with a case study on the Indonesian Red Cross.	Parameters (inputs): - Disaster intensity (Gaussian distribution). - Initial inventory levels at warehouses. - Management strategies for sourcing, transport, replenishment, distribution (traditional vs refined). - Scenario features (encoded strategies, demand, inventory). Performance Metrics (outputs): - Cumulative delivery (pt): percentage of demand met over time. - Response coverage: ability to deliver aid continuously. - Response speed: slope of cumulative delivery curve. - Readiness (r): overall long-term performance metric (0–100%).
33	Inventory Analysis and Optimization	Humanitarian and Emergency	Disaster Relief Inventory Simulation: Managing Resources in Humanitarian Camps	2024	Winter Simulation Conference (WSC)	Analysis: inventory replenishment policies for relief items in camps under perishability, budget limits and uncertain demand	Java	DES	Simulation-based sensitivity analysis	Items, camps and agencies are the core entities, with decision-dependent and system events. Experiments vary replenishment cycle length, funding and buffer ratios. Directly relevant to perishability.	items and shelf life, camps, agencies, demand, replenishment cycle, funding, buffer ratio; performance: unmet demand, expired units, cost
34	Inventory Analysis and Optimization	Industrial and Manufacturing	Dynamic Spare Parts Inventory Management Utilizing Machine Health Data	2024	Winter Simulation Conference (WSC)	Optimization: dynamic spare parts base-stock rules driven by machine health states	Not specified	Other	Continuous-time Markov chain for small instances; simulation-based comparison of dynamic policies	Machines have five health states and workstations have a cold-standby machine. Dynamic rules beat the optimal static base-stock policy by about 7.7%, and the best non-Markovian rule by 18.3%.	number of workstations, machine health transition rates, base-stock level, holding cost, major and minor penalty costs; performance: total average cost

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35	Inventory Analysis and Optimization	Industrial and Manufacturing	Keeping Them Flying: Predicting Spare Parts Inventory for Jet Fighters Using SAS	2024	Winter Simulation Conference (WSC)	Optimization: spare parts inventory levels to reach a target availability at minimum inventory investment	SAS	Monte Carlo	Simulation-based optimization with reliability (lifetime distribution) modelling	Asset lifetimes are estimated from right-censored removal data; the simulation reports mean availability and backlogs with confidence intervals per inventory level. Case description only, with no experimental detail.	asset lifetime distribution, unit cost, order lead time, inventory level per location, budget limit; performance: availability, backlogs, inventory investment
36	Inventory Analysis and Optimization	Other/Not mentioned	Performance of deep reinforcement learning algorithms in two-echelon inventory control systems	2024	International Journal of Production Research (IJPR)	Optimization: production and shipment quantities in a divergent two-echelon inventory system with stochastic, seasonal demand	Python (OpenAI Gym, Ray RLlib and Tune)	Other	Deep reinforcement learning (PPO, A3C, PG) compared with static (s,Q) and other inventory policies	Formulates the problem as a Markov decision process and builds an OpenAI Gym simulator, trained and tuned with Ray RLlib and Tune; static policy parameters are set by Bayesian optimisation on the same simulator. 12 experiments (lead time 1 or 2, one or two product types, one to three local warehouses), each evaluated over 1,000 test episodes against a perfect-information oracle. PPO performs best across experiments; the (s,Q) policy is a solid, cheaper alternative. The paper does not name a simulation paradigm: the simulator is a periodic-review, time-stepped model with sinusoidal demand plus uniform noise, recorded as Other. Code: https://github.com/frenkowski/SCIMAI-Gym .	central warehouse and local warehouses with capacities, number of product types, replenishment lead times, stochastic seasonal demand, holding, backorder and production costs; performance: total cost, training time
37	Inventory Analysis and Optimization	Healthcare and Medical	Simulation- optimisation approach to support management of blood components inventory	2024	Journal of Simulation (JoS)	Optimization: inventory size and replenishment point for red blood cells, balancing shortage and wastage	Arena	DES	Simulation-based optimization (OptQuest)	Blood bank in Campinas, Brazil. One-year runs with a six-month warm-up; about 120 units meets demand with acceptable shortage and wastage. Directly relevant to perishable inventory.	donations, demand, shelf life, maximum inventory, replenishment point, allowed wastage and shortage; performance: expired units,
38	Inventory Analysis and Optimization	Industrial and Manufacturing	Advanced Dynamic Spare Parts Inventory Management Utilizing Machine Health Data	2025	Winter Simulation Conference (WSC)	Optimization: state-dependent spare parts ordering policies for workstations with standby machines, including a two-part extension	Not specified	Other	Markov chain model for small instances; simulation-based comparison of static and dynamic policies with sensitivity analysis	Batch-means output analysis after a warm-up. State-dependent policies save up to 29% against static base-stock policies.	workstations, machine health states and transition rates, base-stock level, order lead time, holding and penalty costs; performance: total average cost, unavoidable cost
39	Inventory Analysis and Optimization; Resilience Modeling, Risk Estimation and Mitigation; Logistics Optimization and Route Planning	Agriculture and Food; Industrial and Manufacturing	An adaptive simulation-based inventory optimisation approach to enhancing biomass supply chain resilience: a case study of remote communities	2025	International Journal of Production Research (IJPR)	Inventory 1. Multi-echelon inventory and replenishment (s, S), 2. Conversion and production processes (at bio-hubs (drying, grinding, pelletizing, cooling) including machine failures and repair times.) 3. Transportation constraints: trucks and ships with seasonal access windows and load capacities	AnyLogic	Hybrid DES + ABS + Monte Carlo	OptQuest	Adaptive inventory planning and resilience in biomass supply chains facing stochastic feedstock quality, seasonal transport windows, equipment breakdowns, and capacity limits. Objective: minimize total system cost while improving reliability, biomass share in the energy mix, and reducing CO2 emissions compared to diesel-only scenarios.	Key modeling parameters Inventory policy variables: reorder point s and order-up-to level S per node. Production/process parameters: dryer heat demand, pellet mill capacity, machine time-to-failure and repair distributions. Feedstock quality: moisture content (Weibull) and heating value (Normal). Transport parameters: truck/ship capacities, speeds, load/unload times, seasonal transport windows. Cost parameters: storage costs (hub and community), transportation costs, electricity leveled costs for biomass and diesel, penalty for diesel use. Performance metrics and outputs Electricity generation mix: total bioenergy (GW) vs diesel (GW); bioenergy share of supply. Total system cost and unit cost of electricity (\$/kWh). CO2 emissions from transportation and fuel combustion (kg CO2-eq). Unmet demand (amount covered by diesel) and reliability/readiness indicators. Inventory dynamics: on-hand and in-process inventory time series at hubs and communities. Cost incurred: inventory, transportation, generation, and module-level cost.

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40	Inventory Analysis and Optimization	Industrial and Manufacturing	Available-to-Promise vs. Order-Up-To: A Discrete-Event Simulation of Semiconductor Inventory Trade-Offs	2025	Winter Simulation Conference (WSC)	Analysis: static and dynamic order-up-to policies against available-to-promise allocation under 26-week lead times	AnyLogic	DES	Simulation-based policy comparison	Ten simulated years on synthetic data regenerated from one year of orders and forecasts. Static order-up-to gives the shortest backlog (9.2 days) at high finished-goods cost; available-to-promise nearly eliminates finished-goods inventory and stays within 22% of its total cost.	fabs, assembly and test, inventory hubs, capacity, 26-week lead time, forecasts and forecast error, safety stock, order-up-to target; performance: backlog duration, backorder share, finished-goods and raw-material inventory cost, total cost, EBIT
41	Inventory Analysis and Optimization	Industrial and Manufacturing	Privacy Meets Performance: Enhancing Distributed Simulation-based Federated Multi-agent Learning with Privacy-preserving Surrogate Model	2025	ACM Transactions on Modeling and Computer Simulation (TOMACS)	Optimization: inventory management policies of SC entities to maximise SC profit while keeping data and simulation models private	Not specified	ABS	Federated multi-agent reinforcement learning with a distributed surrogate model	Extends a privacy-preserving federated multi-agent RL framework built on distributed simulation with a surrogate model that replaces the simulation during training. Evaluated on an aerospace SC; reports higher SC profit for the same number of simulation runs, faster convergence and lower execution time. The paper does not name the simulation paradigm; ABS is inferred from the SC model it reuses (Wang et al. 2022, IJPR, a multi-agent simulation) and from each SC entity being an agent with its own local model.	SC entities as federates each with its own simulation model; inventory decisions; performance: SC profit, training convergence, execution time
42	Inventory Analysis and Optimization; Planning, production planning	Other/Not mentioned	Simulation-Based Online Retailer Supply Chain Inbound Node Arrival Capacity Control	2025	Winter Simulation Conference (WSC)	Optimization: aligning purchase orders with weekly inbound arrival capacity at warehouses while protecting customer service	Not specified	DES	Simulation-based optimization: dual decomposition with a sequential quadratic programming search over capacity costs	Per-product simulations run in parallel and their capacity violations act as a sub-gradient. Capacity adherence improves by over 60% across five weeks with less than 4% loss in a measure of inventory proximity to customers.	products, inbound nodes, weekly arrival capacity, purchase orders, vendor lead times, demand forecasts, inventory placement and transfers; performance: capacity adherence, customer service and inventory proximity measures
43	Inventory Analysis and Optimization	Industrial and Manufacturing	System Dynamics Simulation of Main Material Inventory Control Policy for Electricity Distribution Network Operations	2025	Winter Simulation Conference (WSC)	Analysis: continuous review (s,S) policy, internal coordination and supplier improvement for cables, cubicles, insulators and arresters	Not specified	SD	Simulation-based comparison of eight policy scenarios	Ten sub-models from material request to service duration. Continuous review with supplier improvement performs best: up to 26% budget saving and higher service levels (for power cables, up 26.2%).	material requests, purchase orders, order receipt and lead time, reorder point s and maximum S, supplier delivery timing, budget; performance: service level, inventory turnover, total cost, service duration
44	Inventory Analysis and Optimization; Demand Forecasting	Healthcare and Medical	The effect of visibility on forecast and inventory management performance during the COVID-19 pandemic	2025	International Journal of Production Research (IJPR)	Analysis: effect of delayed, temporally aggregated and erroneous data on forecast accuracy and inventory performance for PPE in a healthcare SC during a pandemic	Python	Monte Carlo	Simulation-based analysis; naive, Holt and epidemiological (SEIRHD) forecasting with a base-stock policy	1,000 Monte Carlo iterations on semi-simulated data across four scenarios. Visibility problems hurt performance in general, but performance improves when they cause large over-forecasts, as with the epidemiological model.	demand, review period, lead time, base-stock level, data delay, aggregation level, reporting error; performance: shortage, inventory cost, forecast error
45	Inventory Analysis and Optimization	Agriculture and Food	The impact of Blockchain adoption on supply chain performance: evidence from food industry	2025	International Journal of Production Research (IJPR)	Analysis: impact of blockchain adoption on inventory level, service level, lead time and operational cost in a food SC	Vensim	SD	Simulation-based scenario analysis	Questionnaire and interview data feed causal loop and stock-flow models. Blockchain improves inventory level and lead time in the short term and cost in the long term, but dropping inventory buffers can reduce the service level.	inventory, orders, lead time, information delay, safety stock; performance: inventory level, service level, lead time, operational cost
46	Inventory Analysis and Optimization	Other/Not mentioned	When Interpretability Meets Efficiency: Integrating Emulation into Supply Chain Simulation	2025	Winter Simulation Conference (WSC)	Simulation efficiency: forecasting inventory flows (order placing, procurement, inbound, outbound) far faster than a full digital-twin simulator	Not specified	Other	Machine-learning emulation inside the simulation; Monte Carlo for vendor arrivals	Runtime falls by orders of magnitude at competitive accuracy. Used for counterfactuals on demand forecasts and vendor lead times and for system interventions on Amazon's SC.	target inventory position, order quantity, purchase orders, vendor lead time distribution, inbound arrivals, outbound demand; performance: inventory prediction accuracy, runtime
47	Inventory Analysis and Optimization	Industrial and Manufacturing	A bio-inspired reorder optimisation framework for lumber supply chain management using a problem-adapted European beaver optimiser	2026	International Journal of Production Research (IJPR)	Optimization: reorder point, order quantity and safety stock in a lumber SC under stochastic demand, disruptions, seasonality and sustainability constraints	MATLAB	ABS	European Beaver Optimiser (bio-inspired metaheuristic), compared with a genetic algorithm and particle swarm optimisation	The optimiser is embedded in the SC simulation and tested in four scenarios; it gives lower median total cost and narrower cost distributions than the baseline policy.	reorder point, order quantity, safety stock, lead time, ordering, holding, shortage and transportation costs, demand variability, supply disruption; performance: total SC cost
48	Inventory Analysis and Optimization; Planning, production planning	Agriculture and Food	A system dynamics-based digital twin application for mealworm farming	2026	Journal of Simulation (JoS)	Analysis: pricing, feed and maturity-cycle policies of a start-up mealworm farm, balancing supply with demand	Vensim; Stella Architect	SD	Simulation-based policy experiments	Built from interviews with a farmer and presented as a digital twin with a user interface. Tests markups for fastest return on investment and the effect of shorter maturity cycles and costlier organic feed.	production stock and maturity delay, inventory, feed cost, price markup, customer demand and market dynamics; performance: return on investment, retained earnings, customer satisfaction

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49	Inventory Analysis and Optimization	Agriculture and Food	Balancing profitability and waste reduction: optimizing markdown policies for retail inventory	2026	International Journal of Production Research (IJPR)	Optimization: markdown (expiration-based discount) policies for perishable retail inventory, trading profit against food waste	AnyLogic	DES	Simulation-based experiments over 830 scenarios	One-year horizon with a 30-day warm-up. Markdown efficiency depends mainly on how likely customers are to pick the freshest item (last-expired-first-out withdrawal); small discounts can cut food waste by up to 80%.	shelf life, freshness levels, replenishment policy, base stock, discount level and start, price elasticity, customer withdrawal behaviour (LEFO/FEFO), profit margin, waste cost; performance: profit, food waste, service level
50	Inventory Analysis and Optimization	Industrial and Manufacturing	Demand variability in engineer-to-order supply chains: insights from a DDMRP case study	2026	International Journal of Production Research (IJPR)	Analysis: demand-driven MRP (DDMRP) against the current push-based replenishment at supplier level in an engineer-to-order SC	MATLAB	Other	Simulation-based comparison and sensitivity analysis	Trace-driven and deterministic, with no replications or confidence intervals (the authors say so). DDMRP cut on-hand inventory by 21.4%, raised fill rate by 7.5 points and cut backorders by 76%.	shared components, buffer zones, average daily usage window, lead time, purchase orders; performance: on-hand inventory, fill rate, backorders, order count
51	Inventory Analysis and Optimization	Industrial and Manufacturing	Endogenous instability: modeling behavioral, structural, and environmental drivers in the coal-power supply chain	2026	Journal of Simulation (JoS)	Analysis: how price expectations, production capacity inertia and seasonal electricity demand drive instability in coal prices, inventories and shutdowns	Not specified	ABS	Simulation-based analysis with Latin-square design and ANOVA	5 episodes of 5,000 daily steps per scenario. Collective price expectations amplify price volatility, and abundant capacity coincides with more frequent shutdowns; supports de-capacity policy.	coal producer capacity and adjustment, power plant ordering rule, target coal-storage days, seasonal demand, network scale, market price formation; performance: price volatility, shutdown frequency, order fulfilment, plant cost, producer profit
52	Inventory Analysis and Optimization	Agriculture and Food	Impact of digital twin adoption on food supply chain performance: a system dynamics approach	2026	International Journal of Production Research (IJPR)	Analysis: impact of digital twin adoption on inventory level, unfulfilled orders, lead time, operational cost and food waste in a food SC	Not specified	SD	Simulation-based scenario analysis	A mental model from practitioner questionnaires and literature is turned into causal loop and stock-flow diagrams. Validation is limited to structural checks and sensitivity analysis because the model is hypothetical (the authors say so). Same group and design as #59.	inventory, orders, delays, information quality, perishability and waste; performance: inventory level, unfulfilled orders, lead time, operational cost, food waste
53	Inventory Analysis and Optimization; Energy and Carbon Cost Modeling and Optimization	Other/Not mentioned	Impacts of incentive-based programmes on supply chains	2026	International Journal of Production Research (IJPR)	Analysis: effect of incentive-based demand response programmes on wholesaler inventory levels, stockouts and lead times, and the response of inventory control policies	Python	Monte Carlo	Simulation-based analysis; flow-shop scheduling solved with CPLEX (DOcplex)	Demand is drawn from four distributions. Load deferral raises average lead time by roughly 75% to 113% and its variance far more, which propagates to wholesaler inventories.	manufacturer, wholesalers, customer demand distribution, order and holding costs, inventory policy, production cycle length, load deferral; performance: inventory level, stockouts, lead time mean and variance
54	Inventory Analysis and Optimization	Industrial and Manufacturing	Information sharing and confidentiality: a federated multi-agent reinforcement learning approach for supply chain coordination	2026	International Journal of Production Research (IJPR)	Optimization: inventory policies and coordination across SC firms while keeping data and simulation models private	Not specified	ABS	Federated multi-agent reinforcement learning (FedAvgRatio), compared with Q-table optimisation and centralised training	Each firm's simulation model runs locally and synchronises through a distributed simulation data bus. Profit is comparable to full-visibility methods and better than traditional optimisation. The paper does not name the simulation paradigm; ABS is inferred from the SC model it reuses (Wang et al. 2022, IJPR, a multi-agent simulation) and from each SC entity being an agent with its own local model. Role of the simulation: it is the MARL training environment and produces all results (profit, cost, convergence), not a validation step. The aerospace SC model is reused from Wang et al. (2022) and the real-network setting from Zhang et al. (2024); the new part is splitting it into federates that sync orders and shipments hourly. 1-minute time step, 7-day decision cycle, 40-day runs with a 10-day warm-up; the distributed and centralised versions give identical outputs.	firms as federates, inventory decisions, demand, synchronisation period; performance: SC profit, training convergence
55	Inventory Analysis and Optimization	Agriculture and Food	Responsiveness-related trade-offs for sustainable operations in perishable goods supply chains	2026	International Journal of Production Research (IJPR)	Analysis: how replenishment policies, SC configuration and perishability shape economic, environmental and social value; effect of lead time, forecast accuracy and SC load	Not specified	SD	Simulation-based analysis with design of experiments	Built on Sterman's stock management and ageing-chain structures with Bass-type demand; replications because period demand is drawn from a normal distribution. No replenishment policy dominates, and fixed minimum-life-on-receipt agreements perform poorly under dynamic conditions. Directly relevant to perishability.	replenishment policy, lead time, forecast accuracy, shelf life and ageing stages, minimum life on receipt, information transparency, demand; performance: sustainable value (economic, environmental, social), waste, service
56	Logistics Optimization and Route Planning; Inventory Analysis and Optimization	Agriculture and Food	Combining the Internet of Things with Simulation-based Optimization to Enhance Logistics in an Agri-Food Supply Chain	2019	Winter Simulation Conference (WSC)	Inventory routing problem (IRP) is addressed (choosing optimum route when transporting items). choose optimum route to minimize routing cost	Not specified	Monte Carlo	(Biased randomized technique with multi-start) use sim to generate initial solution and then iteratively improve it	Real life case study of farm animal feed SC is presented. The feedstock is transported from main warehouse to farms. Every farm has feed keeping tanks/silos which are mounted with sensor to continuously monitor its levels. When delivery trip is made (to deliver at multiple diff farms - like TSP), those farms with full/sufficient levels can be skipped. OR only select and route to those farms which ordered a shipment in such a way that the routing cost is min	inventory levels, inv cost, route cost, inv policy

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57	Logistics Optimization and Route Planning	Other/Not mentioned	Domain-specific Language for modelling and Simulating Actions in Logistics Networks	2019	Winter Simulation Conference (WSC)	No optimization. Authors propose Logistics Assistance System that analyzes logistics net and suggests promising action to take	SimChain	DES	DES is used to model and analyze logistics net do prediction	The paper showcases an example of logistics action that is modeled using proposed domain specific language	No
58	Logistics Optimization and Route Planning	Industrial and Manufacturing	Real-time Supply Chain Simulation: A Big Data-Driven Approach	2019	Winter Simulation Conference (WSC)	optimization is not performed here, rather a framework that can handle large data from Big Data Warehouse to simulate a supply chain is proposed The data contains logistics, production and purchase info	Simio	DES	No optimization is done	Automotive plant supply chain (WSC 2019). Proposes a solution combining a Big Data Warehouse with a Simio simulation model that automatically adapts to the data stored in the warehouse, so the model adjusts to new scenarios without manual intervention and can load the state of the system from the warehouse. Emphasis is on the real-time modelling approach rather than on optimization.	No
59	Logistics Optimization and Route Planning	Other/Not mentioned	Simulation-based Optimization in Transportation and Logistics: Comparing Sample Average Approximation with Simheuristics	2019	Winter Simulation Conference (WSC)	transportation and logistics stochastic team orienteering problem: to find the m visiting routes that maximize the expected aggregated reward	Not specified	Monte Carlo	SBO	A case study addressing stochastic team orienteering problem is presented, two SBO methods are employed (Sample Average Approx SAA and Simheuristics). The problem is represented using a graph	Yes. Nodes(customers), rewards, route(arc) (if present 1, absent 0)
60	Logistics Optimization and Route Planning; Energy and Carbon Cost Modeling and Optimization	Agriculture and Food	Towards A More Sustainable Future? Simulating The Environmental Impact of Online and Offline Grocery Supply Chains	2020	Winter Simulation Conference (WSC)	Analysis analyze the impact of e-grocery SC in terms of mileage and CO2 emission	AnyLogic	DES	Analysis	e-grocery supply chains are modeled and simulated. The performance measures of the simulation are mileage, throughput, utilization etc. It is compared with traditional grocery delivery supply chains Emission model is proposed separately, that computes the emission based on the distance covered and the type of the vehicle used.	Yes. parameter table is presented vehicle type, transport capacity, vehicle utilization, daily delivery freq, daily shopping freq, locations, vehicle speed
61	Logistics Optimization and Route Planning; Energy and Carbon Cost Modeling and Optimization	Other/Not mentioned	Carbon Policies in Network Distribution: A Simulation Approach for Sustainable Supply Chains	2022	Winter Simulation Conference (WSC)	Analysis: how diff carbon policies impact transportation costs and fleet composition	AnyLogic	ABS	Analysis	The paper investigates the impact of carbon policies on supply chain distribution networks. How diff carbon policies impacts distribution costs, fleet composition (omni-channel distribution)	cost calculation policy (dist based costs, and fixed costs - insurance, equipment) carbon policy (carbon tax policy vehicle type (and their carbon emission) Inventory policy (min-max policy)
62	Logistics Optimization and Route Planning; Inventory Analysis and Optimization	Agriculture and Food	Simulation-Based Order Management for the Animal Feed Industry	2022	Winter Simulation Conference (WSC)	Optimization (multiple aspects are modeled) transportation route selection, customer order management to minimize costs	AnyLogic	ABS	Simulation-based and fuzzy based decision strategy	The paper discusses a simulation model for the animal feed industry, focusing on improving order management. It proposes a fuzzy-based decision strategy for customers to decide when to order specific products. (Silo level sensors are used to monitor Silo level)	Specific: num of dairy cattle, hens, silo level, price shift
63	Logistics Optimization and Route Planning; Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	A dynamic and reactive routing protocol for the physical internet network	2024	International Journal of Production Research (IJPR)	Analysis: routing of containers in a Physical Internet sub-network under disruptions (shortages, resource or hub unavailability, emergency and cancelled orders)	AnyLogic	Hybrid DES + ABS	Simulation-based evaluation of a routing protocol with TOPSIS and goal-programming assignment	Proposes a dynamic, reactive routing protocol for the Physical Internet and evaluates it in a multi-agent simulation with order data from two French retail chains and 106 suppliers. 100 replications per experiment; improved robustness and efficiency under disruptions.	facility locations, containers, transport resources, hubs, orders, disruption types; performance: cost, lead time
64	Logistics Optimization and Route Planning	Other/Not mentioned	A Flexible Educational Simulation Model To Study Autonomous Last-Mile Logistics	2024	Winter Simulation Conference (WSC)	Analysis: logistics concepts, fleet configurations and planning algorithms for autonomous last-mile delivery	Plant Simulation; Python	DES	No optimization; a base model for student experiments	Generates the network from OpenStreetMap and builds demand top-down from yearly estimates. Intended as a case-based learning sandbox to evaluate efficiency, emissions and livability.	road network, delivery locations, parcel demand, vehicle fleet, hub; performance: operational efficiency, carbon emissions, livability indicators
65	Logistics Optimization and Route Planning	Other/Not mentioned	Application of microscopic discrete event-based simulation in the modelling of the city logistics systems of concentrated sets of delivery locations	2024	Journal of Simulation (JoS)	Analysis: goods delivery to urban concentrated sets of delivery locations, with and without a consolidation centre	Python	DES	Simulation-based analysis	Describes why earlier Excel and AnyLogic models hit size limits and presents a purpose-built Python DES with pilot models and runtime tests. The model is in a first, pilot phase.	suppliers, stores, consolidation centre, vehicles, delivery transactions, loading bays; performance: number of deliveries, vehicle kilometres,
66	Logistics Optimization and Route Planning	Other/Not mentioned	Assessing the Effects of Container Handling Strategies on Enhancing Freight Throughput	2024	Winter Simulation Conference (WSC)	Analysis and optimization: moving container classification from the port to inland warehouses with intermodal equipment, under workforce shortages	AnyLogic	ABS	Simulation-based parameter variation and optimisation of equipment installation; ANOVA	Nine agent types across 470 locations in four US states. Installing intermodal capability at selected warehouses cuts transport cost, raises throughput and balances resources at the port.	port, warehouses, trains, port and warehouse trucks, orders, workforce availability, intermodal equipment locations; performance: unmet demand, resource utilisation, total cost

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67	Logistics Optimization and Route Planning	Other/Not mentioned	Catalyzing Intelligent Logistics System Simulation with Data-driven Decision Strategies	2024	Winter Simulation Conference (WSC)	Simulation accuracy: emulating operational routing decisions in a large logistics network with data-driven strategies instead of fixed rules	Java; Python	DES	Machine learning model predicts parcel routing decisions inside the simulation	The learned strategies account for resource availability, delivery urgency and operator habits, and give higher simulation accuracy than rule-based logic on real large-scale data.	parcels, sorting centres, routes and next hops, resource availability, shifts; performance: strategy prediction accuracy, simulation accuracy
68	Logistics Optimization and Route Planning	Agriculture and Food	Discrete event simulation as a decision tool: a cost benefit analysis case study	2024	Journal of Simulation (JoS)	Analysis: truck allocation policy, equipment and staffing at a dry bulk (grain) port terminal, and the return on investment of alternatives	Arena	DES	Simulation-based scenario analysis with cost-benefit analysis	The process is described in UML, simulated and validated against the real system; 10 replications per scenario. The rigid 40/60 truck allocation between silos and vessel is the main bottleneck.	truck arrivals, allocation policy, loading equipment, human resources, traffic congestion; performance: waiting and turnaround times, throughput, emissions, investment return
69	Logistics Optimization and Route Planning	Other/Not mentioned	Port Management Digital Twin and Control Tower Integration: an Approach to Support Real-time Decision Making	2024	Winter Simulation Conference (WSC)	Analysis: predictive, real-time decision support for operations at a Brazilian port, looking up to 72 hours ahead	ProModel	DES	Simulation-based prediction and scenario evaluation	The architecture starts the simulation from the current system state and combines digital twin, simulation logic and control tower in one model. Describes specification, development and assisted operation of the project; few quantitative results.	vessel arrivals and departures, berths, tides, handling resources, current system state; performance: predicted operational indicators
70	Logistics Optimization and Route Planning; Supply Chain Design	Other/Not mentioned	PortalLite Towns: Investigating the Viability and Impact of Distributed Small Ports Network in Enhancing Accessibility and Sustainability	2024	Winter Simulation Conference (WSC)	Analysis: cost, efficiency and environmental effect of a distributed small-ports network against centralised direct trucking	AnyLogic	DES	Simulation-based comparison with a mathematical cost model	15 replications of one month, sized from a 30-run pilot. The concept lowers logistics cost in cities with waterways while demand stays below a threshold, and can pay back its investment quickly under good conditions.	primary port, small ports, destinations by population cluster, demand rate, transport costs, infrastructure cost; performance: cost per order, emissions, order fulfilment
71	Logistics Optimization and Route Planning; Planning, production planning	Industrial and Manufacturing	Simulation and Optimization-Based Planning of the Use of Tank Containers in the Production of Specialty Chemicals	2024	Winter Simulation Conference (WSC)	Optimization: which raw materials to supply through tank containers connected to the plant, and the effect on the site logistics system	AnyLogic; Python (Gurobi)	Hybrid DES + ABS	Hybrid: exact optimisation selects materials, simulation evaluates the logistics effect	Two supply scenarios are compared on a real case: manual handling with general cargo containers only, and tank containers added. Parameterisation from a database is automated.	raw materials, tank container storage areas and pipelines, general cargo containers, logistics employees, production demand; performance: employee utilisation, container handling volumes
72	Logistics Optimization and Route Planning; Inventory Analysis and Optimization	Industrial and Manufacturing	Simulation-Based Capital Investment Decision-Making In Steelworks	2024	Winter Simulation Conference (WSC)	Analysis: feasibility of new logistics flows and blending methods, and whether a temporary yard and extra blending-bin capacity are needed	Not specified	Other	Simulation-based scenario analysis	Under a week of severe weather the inventory recovers to steady state within 11 days, so the temporary yard was not built; with random breakdowns for a month the blending-bin inventory stays above its lower limit. Case description with little experimental detail.	ship arrivals, berth, yard, trucks, semi-finished product and ore inventories, blending ratios, weather stoppages, equipment breakdowns; performance: inventory level, logistics cost
73	Logistics Optimization and Route Planning; Energy and Carbon Cost Modeling and Optimization	Agriculture and Food	Sustainable and integrated rail operations planning for grain exports	2024	Journal of Simulation (JoS)	Optimization: assignment of trains to routes to minimise CO2 emissions without reducing exported grain volume	Arena	DES	Simulation-based optimization (OptQuest: scatter search, tabu search, neural network)	Terminals, railways and ports form a closed circuit modelled as a queuing system with 2021 data from Brazil. CO2 could fall by 5.32 kt a year while exports rise 2.20%.	terminals, ports, locomotives, wagons, routes, volumes, storage capacity, supply and demand of grain; performance: CO2 emissions, exported volume
74	Logistics Optimization and Route Planning	Other/Not mentioned	Design Options to Increase the Efficiency of Intra-Port Container Transports	2025	Winter Simulation Conference (WSC)	Analysis: configuration of truck appointment systems (slot length, overbooking, swapping and adding slots, open access, opening hours) for intra-port container transport	Plant Simulation	DES	Simulation-based design of experiments	50 replications per run, no warm-up because truck orders are planned day by day. Medium-length slots work best with transparent processes and long slots with limited transparency; flexibility options improve the share of loaded trips and waiting times.	terminals and other port nodes, trucks, transport orders, time slot length, booking success probability, overbooking, opening hours; performance: waiting time, share of loaded trips, utilisation
75	Logistics Optimization and Route Planning; Planning, production planning	Industrial and Manufacturing	Installation planning for an offshore wind farm: a hybrid modelling framework of integrating simulation and optimisation with a Markov Chain	2025	Journal of Simulation (JoS)	Optimization: vessel loading and installation scheduling to minimise total installation time of an offshore wind farm under uncertain weather	Simio; MATLAB; CPLEX	DES	Hybrid: MILP called inside the DES, with Markov-chain Monte Carlo weather scenarios	Simio triggers the optimiser through MATLAB at events such as a vessel arriving in port. Sensitivity to number of turbines, number of vessels and season is analysed for the Southwest Offshore Wind Farm in South	turbine components, vessels, port, installation sequence, wind speed and wave height states; performance: total project duration
76	Logistics Optimization and Route Planning	Other/Not mentioned	MASS-GT: An empirical model for the simulation of freight policies	2025	Simulation Modelling Practice and Theory (SIMPAT)	Analysis: impact of urban freight policies on shipments, tours, traffic and emissions	Not specified	ABS	Simulation-based policy analysis with discrete choice and optimisation sub-models	Simulates logistic choices of shippers, carriers, producers and consumers: shipment synthesis, vehicle and shipment size, delivery time, tour formation and parcel delivery. Calibrated on truck trip diaries and validated with traffic counts. The model code is open source on GitHub (mass-gt); the paper does not name the language.	firms, shipments, vehicle types, distribution centres, parcel demand, delivery time windows, tours; performance: freight traffic intensity per link, emissions
77	Logistics Optimization and Route Planning	Industrial and Manufacturing	Optimization of Operations in Solid Bulk Port Terminals using Digital Twin	2025	Winter Simulation Conference (WSC)	Optimization: scheduling of railway unloading, stockyard storage and ship loading at a mineral export terminal	AnyLogic; Python	ABS	Ant Colony System metaheuristic with a constructive heuristic, evaluated in the simulation	Validated with real data from a Brazilian terminal. A baseline and three scenarios (direct shipment, preventive maintenance, simultaneous stockyard-to-ship routes) are tested; port stay time is reduced.	trains, stockyard stacks, equipment and routes, ships, maintenance; performance: ship stay time, unloading time

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78	Logistics Optimization and Route Planning	Agriculture and Food	Self-Organization in Crowdsourced Food Delivery Systems	2025	Winter Simulation Conference (WSC)	Analysis: how courier autonomy and cooperation affect delivery efficiency, workload balance and robustness	NetLogo	ABS	Simulation-based experiments	Couriers decide on accepting orders and relocating, using memory of past experience. 30 replications per experiment. Moderate autonomy and cooperation give the best workload distribution and responsiveness.	couriers, restaurants, customers, job generation, autonomy level, cooperativeness level, memory; performance: job waiting time, courier productivity, workload balance
79	Logistics Optimization and Route Planning; Planning, production planning	Healthcare and Medical	Simulation-Based Optimization for CAR T-Cell Therapy Logistics	2025	Winter Simulation Conference (WSC)	Optimization: transport mode selection and patient scheduling in a time-sensitive, centralised-manufacturing therapy SC	SimPy	DES	Simulation-based optimization with two-stage simulated annealing, compared with FIFO, SPT and EDD rules	Patients arrive by a Poisson process; cells travel from hospitals to one manufacturing site with capacity 42 and back. The simheuristic gives lower total cost than the rules across time-window settings while keeping delivery timely.	hospitals, transport modes and costs, manufacturing capacity and process time distributions, patient time windows; performance: total cost, transport cost, delay cost
80	Logistics Optimization and Route Planning	Humanitarian and Emergency	Simulator for cost analysis of inconsistent and unsolicited in-kind donations: insights and projections from a sustainable perspective	2025	International Journal of Production Research (IJPR)	Analysis: economic, social and environmental cost of inconsistent and unsolicited in-kind donations in disaster response under different donation policies	Excel (VBA)	Monte Carlo	Simulation-based policy comparison and sensitivity analysis	Built with Design Science Research on data from the 2022 Petropolis disaster in Brazil and evaluated in eight expert interviews. Problems with donations cost more than six times the minimum achieved by the best policy. Note that it is a cost calculator with random demand rather than a process model.	demand for kits (normal distribution), donated quantities, share of inconsistent donations, transport, storage, deprivation and incineration costs; performance: total cost
81	Logistics Optimization and Route Planning	Other/Not mentioned	A discrete-event simulation framework for smart logistics nodes with agent-based control of hub-to-hub transport	2026	Simulation Modelling Practice and Theory (SIMPAT)	Analysis: effect of connected automated transport on inbound flows and hub-to-hub container transport in smart logistics nodes	Plant Simulation	Hybrid DES + ABS	Simulation-based analysis with planning and control heuristics	Four Dutch logistics nodes are modelled in 3D from open data and validated on road network and freight flows. Connectivity, automation and electrification give 10% to 40% gains in efficiency and emissions.	multimodal network, terminals and warehouses, arrivals by mode, vehicle fleet, parking and decoupling options, charging; performance: logistics efficiency, emissions, waiting times
82	Logistics Optimization and Route Planning	Agriculture and Food	Intelligent scheduling in agricultural wholesale markets: an optimisation approach with simulation-driven and GenAI-based interaction support	2026	International Journal of Production Research (IJPR)	Optimization: truck unloading schedule at an agricultural wholesale market to reduce congestion, waiting and idling emissions	Python 3	DES	Heuristic plus genetic algorithm under fairness constraints (Gurobi for benchmarks), evaluated by simulation	Against first-come-first-served, total unloading time falls by nearly 10% and average waiting by over 20%. A GenAI interface gives instructions to drivers. The paper calls it a discrete-event simulation (Section 2.4), implemented as a fixed time-step loop (Algorithm 1): Poisson truck arrivals, three truck sizes carrying 1-4 product types, unloading time per pallet with a $N(1, 0.1)$ uncertainty factor, limited unloading space, and waiting and priority queues with a skip limit. Scope is inbound logistics at one wholesale market node; upstream arrivals and downstream pickups are exogenous inputs. Gurobi is used only for optimal benchmarks.	truck size, shipment content, unloading duration, unloading space, arrival times; performance: total operation time, waiting time, CO2 from idling
83	Logistics Optimization and Route Planning; Energy and Carbon Cost Modeling and Optimization	Other/Not mentioned	Simulation-based assessment of container-on-barge integration in Texas multimodal logistics	2026	Journal of Simulation (JoS)	Analysis: effect of shifting 5% to 15% of containers to container-on-barge on fuel cost, greenhouse gas emissions and system time	SimPy	DES	Simulation-based scenario analysis	Four scenarios, 10 replications over one year with a 720-hour warm-up. A 10% barge share under congestion cuts fuel cost by up to 6.94% and CO2 by 6.07%; system time rises 5.56% under demand growth.	origin-destination pairs, hubs, vehicle pools by mode, speeds, service times, delays, modal share, demand; performance: fuel cost, CO2, CH4 and N2O emissions, system time
84	Logistics Optimization and Route Planning	Other/Not mentioned	The impact of import container flow characteristics on port operational efficiency	2026	Simulation Modelling Practice and Theory (SIMPAT)	Analysis: how the mix of import container types (standard, hazardous, refrigerated) and sizes affects terminal efficiency	Witness Horizon	DES	Simulation-based scenario and sensitivity analysis	Scenarios come from a synthetic data generator for logistics flows. Container type composition affects dwell time, waiting containers and tractor workload, and a balanced mix is best; container size has a negligible effect.	container type and size shares, storage areas, terminal tractors and other equipment, train and truck departures to hinterland destinations; performance: dwell time, average and maximum waiting containers, equipment utilisation

Sr	Supply chain problem	Application domain	Title	Year	Venue	Specific aspect / SC aspect	Tools used to build the model	Simulation method used	Optimization/ prediction method used	Comments	Any attributes
85	Planning, production planning	Industrial and Manufacturing	Semiconductor wafer fabrication production planning using multi-fidelity simulation optimisation	2020	International Journal of Production Research (IJPR)	Planning Production planning	Not specified	DES	Simulation-based optimization	Production/release planning for wafer fabs where cycle times depend on workload (re-entrant flows, batch processing, machine unreliability). The paper tackles the tension between accuracy (high-fidelity DES) and computational cost by proposing a multi-fidelity simulation-optimization approach to select release plans that maximize profit while keeping simulation budget feasible. Problem: Find profitable weekly release plans for a complex, re-entrant wafer fabrication system where cycle times depend on workload, but exhaustive high-fidelity simulation of many candidate plans is computationally prohibitive. How they addressed it: The authors built a fast low-fidelity epoch/queueing approximation that reliably ranks candidate release plans by ordinal performance, then applied the MO2TOS multi-fidelity optimization (Ordinal Transformation + OCBA sampling) to allocate a limited high-fidelity DES budget to the most promising groups. Experiments (70% and 90% utilization scenarios) show MO2TOS finds better release plans faster than SPSA and naive sampling, reducing computation time while improving profit and WIP matching	Parameters: Processing time distributions (lognormal processing times per workstation) Failure/repair distributions (gamma distributions for unreliable machines) Batch sizes Planning horizon (26 weekly periods + extra periods to finish in-process lots) Release bounds Demand Performance metrics and outputs total profit (revenue ? material ? WIP ? inventory ? backlog costs). Cost breakdowns: inventory cost, WIP holding cost, backlog cost, material cost. computational time (total simulation hours), convergence behavior.
86	Planning, production planning	Industrial and Manufacturing	A two-level optimisation-simulation method for production planning and scheduling: the industrial case of a human–robot collaborative assembly line	2022	International Journal of Production Research (IJPR)	Planning Production planning	Simio	DES	Simulation-based optimization	Problem: Produce capacity-feasible, cost-effective production plans and schedules for a human–robot collaborative assembly line where aggregated optimization alone can overestimate capacity and ignore robot travel/setup dynamics. How addressed: The authors developed a two-level MILP (planning + scheduling) embedded in an iterative ROSA loop with a detailed SIMIO DES model. The optimization proposes release quantities, resource allocations and sequences; the simulation validates feasibility, measures realistic utilization/downtime/travel, and returns recursive parameters (λ, ϵ, β, ϑ) that tighten optimization constraints. Iterations continue until convergence (stable total output, late-order rates). Experiments include layout comparisons, stochastic process characterization, and sensitivity analyses showing the method converges to near-optimal, implementable plans while revealing managerial insights on robot allocation and layout design.	Parameters: Resource limits: up to 5 shared robots (minimum 2 per period in experiments); robot speed, setup/changeover times, workspace distances. Processing times Costs: production cost, inventory cost, backlog cost, robot assignment and allocation costs (numerical values used in experiments). Performance metrics: Total operational cost Makespan Number of robot tasks allocated Workstation utilization rates Percentage of late orders / on-time delivery
87	Planning, production planning; Logistics Optimization and Route Planning	Industrial and Manufacturing	Analyzing Delivery Performance and Robustness Of Wood Supply Chains Using Simulation-Based Multi-Objective Optimization	2024	Winter Simulation Conference (WSC)	Optimization: harvest scheduling to maximise service level and throughput while minimising lead time and delivery deviation under weather-dependent road accessibility	FACTS Analyzer	DES	Simulation-based multi-objective optimization (genetic algorithm with partially-mapped crossover)	Population of 50 over 100 generations with ten replications per evaluation (about 24 hours). A loss function adds robustness of deliveries as an objective; Pareto solutions are inspected through a dashboard.	harvest areas and schedule, road accessibility, transport, customer demand; performance: service level, throughput, lead time, delivery deviation (loss function), backorders
88	Planning, production planning; Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	Deploying hybrid modelling to support the development of a digital twin for supply chain master planning under disruptions	2024	International Journal of Production Research (IJPR)	Optimization: production, storage and distribution quantities of a SC master plan under demand and lead time disruptions	Python (SimPy)	DES	Hybrid: simulation + decision tree algorithms + optimisation (Pyomo, CPLEX)	Hybrid modelling framework presented as a precursor to a SC digital twin. Tested on demand increase and lead time extension scenarios with 100 replications; finds it is more economical to hold higher inventory downstream than upstream under disruptions. The paper states it applies the DES methodology, built with the SimPy library (written "Simply" in the paper), with a 1-week time step over 52 weeks and a 12-week warm-up.	demand, lead time, production, storage and distribution quantities, inventory at each SC member; performance: service level, total cost
89	Planning, production planning; Logistics Optimization and Route Planning	Industrial and Manufacturing	Discrete Events Simulation of a Manufacturing Line for Floating Wind Turbines	2024	Winter Simulation Conference (WSC)	Analysis and optimization: line capacity, procurement schedule and buffer sizes for a 50-unit production programme under weather interruptions	FlexSim	DES	Simulation-based experiments (a first optimisation run)	Three floater workstations, two turbine assembly stations and inland pre-assembly stations; wind speed and tide height interrupt some stations. Better shipment and storage planning reduces expected project duration. Work in progress.	workstations, task durations, supplier port and vessel trips, component buffers, weather and tide conditions; performance: project duration, buffer occupancy

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90	Planning, production planning	Industrial and Manufacturing	Long-term Rapid Scenario Planning in the Semiconductor Industry using Deep Reinforcement Learning	2024	Winter Simulation Conference (WSC)	Optimization: updating long-term production plans (allocating capacity to customer orders by priority) after unforeseen events	Not specified	Other	Deep reinforcement learning (Deep Controlled Learning, PPO) compared with a mathematical programming model and dispatching prioritisation rules	DRL gives close-to-optimal plans far faster than the mathematical programme and adapts to different order books, but does not consistently beat tailored dispatching rules. The simulation paradigm is not named.	customer orders with priorities and due dates, resource capacity in hours, 52-week horizon, order book randomness; performance: weighted delivery performance, runtime
91	Planning, production planning; Inventory Analysis and Optimization	Industrial and Manufacturing	Supply Chain Digital Twin Framework for Hybrid Manufacturing Strategy Selection: a Case Study from the Semiconductor Industry	2024	Winter Simulation Conference (WSC)	Optimization: choosing make-to-stock or make-to-order for each product in a manufacturing SC	Not specified	Other	Simulation-based optimization with a genetic algorithm	The framework has a data infrastructure, an online simulation core that updates itself from MRP, demand, cost, production status and stock data, and a decision-making core. Described in BPMN. The simulation paradigm is not named.	products, demand, MRP data, production status, stock levels, costs and revenue; performance: cost and service measures of the manufacturing strategy
92	Planning, production planning; Inventory Analysis and Optimization	Agriculture and Food	A Hybrid Simulation-based Approach for Adaptive Production and Demand Management in Competitive Markets	2025	Winter Simulation Conference (WSC)	Analysis and optimization: production and inventory planning together with demand growth under marketing strategies	AnyLogic	Hybrid DES + SD	Simulation-based experimentation and parameter optimisation; a dynamic marketing strategy algorithm	DES covers order arrivals, production cycles and restocking; SD uses a Bass diffusion model for purchase and replacement demand. Data from a manufacturer in Bangladesh. The marketing algorithm gives about 38% more demand growth.	production resources, stock, order arrivals, restocking, coefficients of innovation and imitation, replacement demand, marketing effort; performance: demand growth, inventory, profit
93	Planning, production planning; Inventory Analysis and Optimization	Healthcare and Medical	Deep reinforcement learning approach for dynamic capacity planning in decentralised regenerative medicine supply chains	2025	International Journal of Production Research (IJPR)	Optimization: dynamic capacity planning (inventory replenishment, capacity allocation, demand sharing) in a decentralised personalised regenerative medicine SC under supplier disruption and demand uncertainty	Python 3.6 (OpenAI Gym; PyTorch for the DDPG agent)	Other	Deep reinforcement learning (DDPG, PyTorch) trained on the simulation	Two manufacturing facilities and a dedicated reagent supplier are simulated; the DRL policy outperforms approximate dynamic programming methods when true demand differs from prior estimates. The authors build the simulation environment themselves in OpenAI Gym (Section 5.1, 'we now construct a simulation environment'): two facilities and a reagent supplier, weekly epochs over 52 weeks, Poisson demand, Bernoulli supplier disruptions, 3-epoch manufacturing time; 500 simulations per evaluation. The problem formulation generalises the capacity planning model of Li and White (2022), whose ADP heuristics are the benchmarks. The paradigm is not named; it is a time-stepped state-transition simulation, recorded as Other (as #25).	manufacturing facilities, supplier, reagent inventory, capacity, demand forecast error, supplier disruption; performance: total production cost, service level
94	Planning, production planning	Healthcare and Medical	Designing Scalable Cell Therapy Processes: a Framework for Simulation, Bottleneck Analysis, and Scale-up Evaluation	2025	Winter Simulation Conference (WSC)	Analysis: bottlenecks, capacity and staffing when scaling up autologous cell therapy manufacturing	FlexSim	DES	Simulation-based design of experiments and sensitivity analysis; optional simulation-optimisation	Four layers: process definition, resources, control policy, and scenario analytics. Balanced scaling of resources and staffing gives larger gains than adding resources alone. Framework description with little quantitative detail.	unit operations and durations, operators, biosafety cabinets, incubators, clean-room slots, shift patterns, batch arrivals, failure and hold events; performance: throughput, lead time, utilisation, queue time, WIP
95	Planning, production planning; Inventory Analysis and Optimization	Industrial and Manufacturing	Digital Twins for Optimizing the Transition from Job-shop to Mass Production: Insights from Marine Pump Manufacturing in Scandinavia	2025	Winter Simulation Conference (WSC)	Optimization: redundancy allocation, pull-order batch sizes and buffer inventory policy when moving from job-shop to mass production	Plant Simulation	DES	Simulation-based optimization with a genetic algorithm; redundancy analysis	A five-step framework on a Norwegian case. The baseline model is validated against ERP cycle times. Cycle time falls from more than a week to about one day and annual capacity exceeds projected demand.	stations, redundancy levels, batch sizes, buffer stock and reorder policy, processing times; performance: cycle time, annual capacity, utilisation
96	Planning, production planning; Demand Forecasting	Industrial and Manufacturing	How periodic forecast updates influence MRP planning parameters: A simulation study	2025	Simulation Modelling Practice and Theory (SIMPAT)	Optimization: MRP planning parameters (lot size, planned lead time, safety stock) under periodic customer forecast updates and forecast bias	AnyLogic	DES	Simulation-based parameter search; a safety stock exploitation heuristic for MRP	Large experiment parallelised over 21 computers. Forecast updates create nervousness in standard MRP; the heuristic lowers inventory plus backorder cost, and fixed-order-quantity beats fixed-order-period in most cases.	lot-sizing policy and lot size, planned lead time, safety stock, forecast uncertainty and bias, forecast horizon, utilisation; performance: inventory cost, backorder cost, overall cost
97	Planning, production planning	Industrial and Manufacturing	Improving Plan Stability in Semiconductor Manufacturing through Stochastic Optimization: a Case Study	2025	Winter Simulation Conference (WSC)	Optimization: master planning (wafer starts) that reduces plan nervousness under demand uncertainty	Not specified	DES	Two-stage stochastic programming against linear programming (Gurobi), evaluated by simulation	Demand arrivals follow a Markov chain and demand sizes a Gamma distribution fitted to history. Stochastic programming gives more stable plans with comparable on-time delivery and inventory position. Optimisation is the core and the simulation is the evaluation bed, as in #27.	products, forecast and sampled demand scenarios, front-end lead time, capacity, replanning period; performance: plan stability, on-time delivery, inventory position
98	Planning, production planning	Industrial and Manufacturing	Improving Plan Stability in Semiconductor Master Planning through Stochastic Optimization and Simulation	2025	Winter Simulation Conference (WSC)	Optimization: master planning that reduces plan nervousness in a semiconductor SC	Not specified	DES	Two-stage stochastic programming against linear programming, evaluated by	Summary of #227 with one demand scenario for LP and five for stochastic programming.	as #227

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99	Planning, production planning; Inventory Analysis and Optimization	Industrial and Manufacturing	Machine learning-based dynamic production planning and control in unreliable manufacturing systems with supply disruptions	2025	International Journal of Production Research (IJPR)	Optimization: integrated production rate (hedging point) and raw material replenishment (order size and trigger) under random failures and delivery lead time variation	Arena	DES	Machine learning (neural network, random forest) with design of experiments and simulation-based optimization	A combined discrete-continuous model written in SIMAN within Arena (continuous inventory levels and production flow; discrete failures, repairs and order arrivals) is run to steady state; an ML cost model then adjusts policy parameters dynamically. Dynamic policies cut cost by up to 15% against semi-dynamic and 25% against static policies when lead time variability is high.	machine failure and repair times, production rate, raw material delivery lead time, order size, reorder trigger, holding, backlog, ordering and production costs; performance: long-run total cost
100	Planning, production planning	Industrial and Manufacturing	Simulating Front-End Semiconductor Supply Chains to assess Master Plans under Uncertainty: a Case Study	2025	Winter Simulation Conference (WSC)	Analysis: performance of master plans under demand and supply uncertainty compared with deterministic expectations	Not specified	DES	Simulation-based analysis	Cycle times come from historical distributions with a load-dependent regression; demand is forecast adjusted by sampled forecast errors. 500 replications. Uncertainty changes on-time delivery and inventory markedly against the deterministic plan.	wafer fabs, bottleneck resources, cycle time distribution, work in progress, forecasts and forecast percentage errors, master plan; performance: on-time delivery, inventory on hand
101	Planning, production planning	Industrial and Manufacturing	Simulation-based Production Planning For An Electronic Manufacturing Service Provider Using Collaborative Planning	2025	Winter Simulation Conference (WSC)	Analysis: collaborative planning (using customer inventory and order data) against isolated planning based on own forecasts	Not specified	DES	Simulation-based scenario comparison	Built on empirical data from one provider and an earlier manufacturing model. In every scenario the collaborative model lowers average inventory while holding the target service level.	components with long and uncertain procurement lead times, customer orders and inventory, forecasts, production process; performance: service level, average inventory
102	Planning, production planning; Inventory Analysis and Optimization	Agriculture and Food	Strategic Production Transformation Through Simulation- Driven Decision Support	2025	Winter Simulation Conference (WSC)	Analysis: operational and economic effect of moving a production line from two to three products, and of storage expansion	Not specified	DES	Simulation-based scenario analysis	More changeovers shorten runs and outbound storage nears its limit, raising stockout risk; shifting volume to another plant and partial or full storage expansion are evaluated. Case description without numerical results.	products, changeover downtime and cleaning, inbound and outbound storage capacities, loading and unloading rates, outbound demand variability; performance: throughput, stockout risk, storage utilisation, cost
103	Planning, production planning; Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Zero-inventory plans, constant workforce, or hybrid approach? Analysing pure production strategies for enhancing factory resilience with demand variability	2025	International Journal of Production Research (IJPR)	Analysis: resilience of a factory under zero-inventory, constant-workforce and hybrid aggregate production planning strategies	@RISK	Monte Carlo	Simulation-based analysis with a heuristic for the pure strategies and a Cobb-Douglas resilience index	10,000 runs over a 36-week horizon with seasonal stochastic demand from a Holt model. Zero-inventory (chase) plans give the highest factory resilience.	weekly demand, production quantity, workforce, inventory, backorders; performance: factory resilience index, cost
104	Planning, production planning; Inventory Analysis and Optimization	Industrial and Manufacturing	Simulation-based planning system for a power to methanol supply chain	2026	International Journal of Production Research (IJPR)	Optimization: production planning, inventory control and logistics coordination strategies in a power-to-methanol SC, assessed techno-economically	AnyLogic; DWSIM	Hybrid DES + ABS	Simulation-based optimization: branch-and-bound-inspired pruning over policy and design combinations; model predictive control and Silver-Meal heuristic for hydrogen production	Steady-state process models (DWSIM) supply parameters to the AnyLogic model, where CO2 suppliers, methanol distributors and logistics providers are coordinated by agent-based planning logic (MPS to MRP). Danish case over 30 years; electricity price variability dominates profitability.	H2, CO2 and methanol inventories, lot-sizing rule, procurement policy, review period, transport capacity, electricity price, solar availability, electrolyser degradation, demand; performance: net present value, internal rate of return, break-even period
105	Resilience Modeling, Risk Estimation and Mitigation; Inventory Analysis and Optimization	Industrial and Manufacturing	Analyzing Pre-Positioning within a Disrupted Bulk Petroleum Supply Chain	2019	Winter Simulation Conference (WSC)	Analysis: supply chain disruption events are modeled. (hurricane) authors model different disruption mitigation strategies and assess their effectiveness	Not specified	Other	Analysis: pre-positioning disruption mitigation strategy is illustrated in paper - leads to optimization problems	authors also show effects of disruption on inventory levels, and resumption of normal inventory conduct after the end of disruption	inventory Policy, inventory params some specific about disruption duration, duration to recover from disruption and so on
106	Resilience Modeling, Risk Estimation and Mitigation	Information and Communication Technology (ICT)	Product Life Cycle Perspective on ICT Product Supply Chain Resilience	2020	Winter Simulation Conference (WSC)	Analysis analyzing the relations between the supply chain configuration and its ability to deal with vulnerabilities.	Not specified	Monte Carlo	Analysis	Information and Telecommunication Technologies (ICT) supply chains, (example: smart watch, vulnerabilities: regular security updates) Authors propose to model and simulate the ICT supply chains to find relationship between SC configuration, vulnerabilities and SC resilience	Yes. product life cycle length, product monitor cost, node failure prob, cost of treating vulnerability
107	Resilience Modeling, Risk Estimation and Mitigation	Agriculture and Food	Simulation-based Digital Twin Development for Blockchain Enabled End-to-end Industrial Hemp Supply Chain Risk Management	2020	Winter Simulation Conference (WSC)	Analysis of Hemp supply chain for risk control	Not specified	Other	Analysis	Hemp is a plant, Industrial Hemp SC includes entire process of planting Hemp seeds, growing them to the final sale. Authors propose to develop Blockchain enabled IHSC to track data and key info, improve transparency and involve local authorities (quality control). They develop a DT to simulate and analyze IHSC to understand the processes in the SC and minimize risks. (Simulation for seed selection)	Yes. Arrival of seeds, pre-plant prepare time, cultivation time, harvest time ... (specific params)

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108	Resilience Modeling, Risk Estimation and Mitigation	Healthcare and Medical	Assessing Resilience of Medicine Supply Chain Networks to Disruptions: A Proposed Hybrid Simulation modelling Framework	2021	Winter Simulation Conference (WSC)	Analysis	AnyLogic	Hybrid DES + ABS + SD	Analysis	Authors model and simulate a medicine SC (wholesaler, warehouse, point of sale, supplier, pharma ingredients producer) with disruptions (demand shock, and supply shock) to improve medicine SC.	Inventory replenishment policy (s,S), inventory parameters
109	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Simulating and evaluating supply chain disruptions along an end-to-end semiconductor automotive supply chain	2021	Winter Simulation Conference (WSC)	Analysis: multi echelon supply chain analysis under disruption events (pandemic/covid)	AnyLogic	Hybrid DES + ABS + SD	Analysis	a simulation model to understand how a multi-level supply chain responds in pandemic. Pandemic causes fluctuations in demand, which leads to operational challenges in SC. To face these, SC partners should work in collaboration, which makes SC more resilient	Not explicitly mentioned
110	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	A System Dynamics Model for Studying the Resiliency of Supply Chains and Informing Mitigation Policies for Responding to Disruptions	2022	Winter Simulation Conference (WSC)	Analysis: study economic impact of supply chain disruptions	Stella Architect	SD	Analysis: hybrid SD and Input Output IO model is developed to study the economic impact of supply chains disruptions	Authors develop SD and IO model of supply chains with disruptions to assess the resilience and understand its economic impact (changes in economy of region or country - employment, gross income, production, etc)	Yes, specific to disruption modeling, speed of inv adjustment, labor hiring adjustment, consumption adjustment
111	Resilience Modeling, Risk Estimation and Mitigation; Inventory Analysis and Optimization; Planning, production planning	Healthcare and Medical	Simulation of IT Data Integration to Optimize an Antibiotics Supply Chain with System Dynamics	2022	Winter Simulation Conference (WSC)	Optimization of Antibiotic Supply chains (Mitigating disruption by improving communications)	Vensim	SD	Simulation-based with Data Integration	authors model on multiple aspects of AB supply chain: planning, reducing lead time, and inventory to prevent shortage and excess. They also discuss data integration for communication: electronic data exchange, point of sale data	shipment rate, production rate, order fulfillment rate, delivery rate, order rate, usage rate (raw product)
112	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	A generic representation of supply network resilience using simulation based experimentation	2023	Journal of Simulation (JoS)	SC network resilience	NetLogo	ABS	Simulation-based	Sector not specified (generic representation). Presents a generic representation of supply network resilience implemented with agent-based modelling and simulation, adapting the SCOR model (Plan, Source, Make, Deliver, Enable) for both the processes and the performance metrics, and parameterised by a firm's sourcing strategy and its product's bill of materials. Demonstrated on single- and dual-sourcing network structures under disruptions of operational and strategic nature.	sourcing strategy (single vs dual sourcing); bill of materials / production vector (technological assembly order); multi-echelon network structure; disruption type (exogenous - infrequent but long duration; endogenous / operational - higher frequency, short duration), intensity and duration performance: SCOR Level-I metrics capturing the rate of response (recovery plus readiness) of the network, e.g. order fulfilment cycle time
113	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	A Hybrid System Dynamics/Input-Output Model for Studying the Impact of Transportation Delays on the Resilience of National Supply Chains	2023	Winter Simulation Conference (WSC)	Analysis: effect of transportation delays on supply chains	Not specified	SD	Analysis	Authors model transportation delays (and flow of goods) in national supply chain (that is responsible for delivering to different industrial supply chains) with disruptions to study its impact on industry and assess resilience	Demand, total production, industry consumption, industry inventory levels, delivery delay
114	Resilience Modeling, Risk Estimation and Mitigation	Healthcare and Medical	A Simulation-Based Method for Analyzing Supply Chain Vulnerability Under Pandemic: A Special Focus on the Covid-19	2023	Winter Simulation Conference (WSC)	Analysis	Not specified	ABS	Analysis: simulation-based	The article presents a simulation-based method to analyze supply chain vulnerabilities during the COVID-19 pandemic. It combines the susceptible-infected-recovered (SIR) model with agent-based simulation to study risk propagation from distribution centers to retailers	Quarantine Start Day: The day quarantine measures begin. Quarantine Duration: The length of the quarantine period. Daily Testing Capacity: The number of tests conducted daily. Contact Tracing Effectiveness: The efficiency of tracing contacts of infected individuals. Symptomatic Test Rate: The rate at which symptomatic individuals are tested. Infectivity: The rate at which the virus spreads.

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115	Resilience Modeling, Risk Estimation and Mitigation; Supply Chain Design	Other/Not mentioned	A Simulation-Optimization Approach for Designing Resilient Hyperconnected Physical Internet Supply Chains	2023	Winter Simulation Conference (WSC)	Optimization: of cost and resilience of supply chain by integrating with Physical Internet	Not specified	DES	Simulation-optimization hybridizing a multi-objective, multi-period mixed-integer program with discrete-event simulation: a baseline (B) MIP and a hyperconnected (H) MIP minimize cost, a further MIP maximizes resilience for the configuration under a simulated disruption, and the loop repeats over simulation runs and over an increasing number of open pi-hubs	Authors propose hybrid model DES+MIP to optimize SC resilience and costs. They model SC operations under disruptions. The Physical Internet encapsulates products in standardized, modular pi-containers routed through pi-hubs. A network hyperconnection strategy is tested: hyperconnecting the supply chain improves both resilience and risk, especially when active edges are disturbed, but incurs higher costs.	SC nodes: supplier, customers, hubs costs capacities (supplier, hubs) Disruptions (and their prob) design decisions: which pi-hubs to open, which edges to activate, and the pi-container flow on each active edge; distance limit per edge; hyperconnection strategy; 26 designed test instances performance (KPIs): total cost (fixed and variable pi-container costs plus distance-related edge costs), resilience, risk, pi-hub utilization, distance travelled by the pi-containers
116	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	Additive manufacturing integration in E-commerce supply chain network to improve resilience and competitiveness	2023	Simulation Modelling Practice and Theory (SIMPAT)	Analysis two echelon SC net with AM	Arena	DES	OptQuest	The paper explores the integration of additive manufacturing (AM), specifically 3D printing, into e-commerce supply chains to enhance resilience and competitiveness. It investigates three utilization policies: reactive, proactive, and no-AM, within a two-echelon supply chain network. The study uses simulation optimization to analyze the impact of AM on various performance metrics like total network cost and response time. Findings indicate that the reactive policy significantly improves network performance, especially under low-demand scenarios, by reducing holding and ordering costs. The research provides valuable insights for managers on optimizing supply chain operations using AM, particularly during disruptions like the COVID-19 pandemic.	production cost, lead time, max response time (mrt), demand distribution, inventory holding cost, ordering cost total sc cost, response time, inventory levels
117	Resilience Modeling, Risk Estimation and Mitigation	Healthcare and Medical; Humanitarian and Emergency	An Integrated System Dynamics and Discrete Event Supply Chain Simulation Framework for Supply Chain Resilience with Non-stationary Pandemic Demand	2023	Winter Simulation Conference (WSC)	Analysis: Hybrid simulation framework to improve resilience	AnyLogic	Hybrid DES + SD	Analysis	The paper proposes a hybrid simulation framework integrating system dynamics and discrete event simulation to enhance supply chain resilience during disruptions like the COVID-19 pandemic Authors present case study of Oxygen Concentrators (medical device used to deliver oxygen to patient) during COVID19 pandemic	Specific: illness duration, hospitalization rate, OC inventory rate, OC inventory levels
118	Resilience Modeling, Risk Estimation and Mitigation	Information and Communication Technology (ICT)	Evaluating the Effectiveness of Countermeasures in ICT Supply Chains through Elicitation-Informed Simulation	2023	Winter Simulation Conference (WSC)	Analysis: evaluating countermeasures (CMs) in Information and Communication Technology (ICT) supply chains (SCs) to combat counterfeiting	AnyLogic	Hybrid DES + ABS	Analysis	authors model ICT (Info Comm Tech) supply chains with disruptions to combat counterfeiting. They also assess the resilience of the supply chain.	hybrid model in which each facility agent (suppliers, assembly, D distribution centres, C customers) embeds discrete-event modules with periodically triggered event blocks; laptops as the modelled ICT product; counterfeiting as the disruption; Delphi elicitation of subject matter experts mapped into input empirical distributions of risk reduction; risk-reduction factor and benchmark risk for the threat-only scenario compared against countermeasure scenarios; key parameters varied in parameter-variation experiments performance: system service level, delivery time against the estimated delivery time, and product quality (authenticity of the delivered products)
119	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	Modeling Risk Prioritization of a Manufacturing Supply Chain using Discrete Event Simulation	2023	Winter Simulation Conference (WSC)	Analysis: Authors build the model to study risks in supply chain, prioritize them and help in decision making and developing mitigation strategies	Python	DES	Analysis: simulation-based	Authors build a simulation model with risks to assess the performance and impact of various risks and prioritize them.	likelihood of risk event, impact, time to fix disruption

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120	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	System Dynamics Simulation of External Supply Chain Disruptions on a Simplified Semiconductor Supply Chain	2023	Winter Simulation Conference (WSC)	Analysis: the impact of external disruptions on the semiconductor supply chain	AnyLogic	SD	Simulation-based	The paper discusses the use of system dynamics simulation to analyze the impact of external disruptions (natural disasters, pandemics etc) on the semiconductor supply chain. Authors build a semiconductor supply chain model, which can be subjected to what-if scenarios to assess the resilience.	Specific parameters to semiconductor supply chains.
121	Resilience Modeling, Risk Estimation and Mitigation; Energy and Carbon Cost Modeling and Optimization	Agriculture and Food	A simulation model for addressing supply chain disruptions under a multi-capital sustainability perspective: a case study in the agri-food sector	2024	Journal of Simulation (JoS)	Analysis of supply chain under disruption; risk analysis	AnyLogistix	DES	Simulation-based	Agri-food sector, supermarket supply chain case study (mozzarella; 7 factories, 1 distribution centre and 9 supermarkets, Italy). Simulates disruptions - an unexpected increase in product demand, facility closure and labour shortage in transportation - and shows that disruptions can be exploited to revitalise small local suppliers while simultaneously improving several sustainability indicators from a multi-capital perspective.	case: mozzarella supply chain with 7 factories, 1 DC and 9 supermarkets; disruptions: unexpected increase in product demand, facility closure, labour shortage in transportation; scenarios SS1-SS4 (closure of a single global supplier) and SM1-SM7 (simultaneous closure of multiple global suppliers) performance (multi-capital sustainability indicators): financial capital (revenue from supermarkets; revenue minus transportation and inventory cost), natural capital (CO2 emissions from vehicles), image capital (CO2 emissions per unit revenue; travelled distance per product in successful orders), stakeholders-related capital (share of successful orders; share of products in successful orders; 1 minus lost revenue over potential revenue), external social capital (share of revenue from local suppliers; share of active local suppliers)
122	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing; Information and Communication Technology (ICT); Healthcare and Medical	A system dynamics model for assessing the ripple effect of intertwined supply networks	2024	Journal of Simulation (JoS)	risk analysis	Vensim	SD	Simulation-based	Semiconductor supply chain and the supply chains it feeds. System dynamics model, built from a causal loop diagram, of the ripple effect in an intertwined supply network: it demonstrates how semiconductor shortages propagate to the automobile, consumer electronics and medical devices supply chains, with a significant impact on automobile firms caused by the demand surge in consumer electronics. Uses secondary data from a UK-based company and argues for stress testing and for adding risk management mechanisms to the list of performance measures.	input conditions varied in the experiments: sudden demand increase in semiconductor applications and capacity disruption at various supply chain nodes; interacting variables of the causal loop diagram; intertwined supply chains modelled: automobile, consumer electronics, medical devices performance: vulnerability index at different time intervals; customer, financial and operational KPIs
123	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	A viable supply chain model for managing panic-buying related challenges: lessons learned from the COVID-19 pandemic	2024	International Journal of Production Research (IJPR)	Analysis: impact of panic buying during a large-scale disruption and effectiveness of recovery strategies and plans	AnyLogic	ABS	Integrated agent-based model and optimisation	Develops a viable SC model with an integrated agent-based and optimisation method, tests four strategies and three recovery plans. Finds that raising production at decentralised facilities, ordering more frequently from multiple suppliers and using third-party transporters mitigate panic buying.	suppliers, manufacturers, retailers, transporters, order frequency, production capacity, demand surge; performance: output, product availability
124	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	An agent-based model of supply chain financing: agents' strategies, performance, and risk contagions	2024	Journal of Simulation (JoS)	risk analysis	NetLogo	ABS	Simulation-based	Supply chain finance (SCF), with the Chinese market used to proxy the key model parameters. Agent-based model providing a comprehensive framework that integrates banks, large businesses and SMEs. Banks prefer large businesses, but trade credit and partial credit guarantee remain a pair of complementary financing channels for SMEs; cautious agent strategies significantly decrease risk contagion in the supply chain, although marketisation negatively moderates this effect for SMEs; longer-term bank credit policies bring lower risk contagion but increase the uncertainty of the banks' risk.	agents: banks, large businesses (core enterprises) and SMEs; agents' strategies (cautious vs less cautious); financing channels: trade credit financing (TF) and partial credit guarantee financing (PF); bank credit policy term; degree of marketisation / market-oriented financial system; key parameters proxied from the Chinese market performance: financing accessibility for SMEs versus large businesses, risk contagion across the supply chain, uncertainty and fluctuation of the banks' risk

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125	Resilience Modeling, Risk Estimation and Mitigation	Agriculture and Food	Assessing operational resilience in food supply chains: a discrete-event simulation approach based on a framework for integrated decision-making	2024	Journal of Simulation (JoS)	resilience analysis	AnyLogic	DES	Simulation-based	Austrian dairy supply chain. Assesses operational resilience during sudden demand spikes using a framework for integrated decision-making that spans strategic, tactical and operational perspectives, with modular building blocks for food supply chain modelling. Upstream buffer stocks, adapted reordering strategies and operational adjustments are all effective, but no single intervention consistently outperforms the others across all performance metrics, so interventions must be selected case by case.	actors: farms, processors, distributors, retailers; interventions by decision level - strategic (upstream buffer stocks), tactical (adapted reordering strategies), operational (operational adjustments); base case of a 50% demand spike lasting two weeks; sensitivity analyses for robustness performance (KPIs): time to recovery (TTR), DTI, share of unmet demand overall, during and outside the disruption and its maximum (SUD, SUDDISR, SUDNDISR, SUMax), number of days at 90%, 95% and 99% service level (SL90, SL95, SL99)
126	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Effects of adaptive cooperation among heterogeneous manufacturers on supply chain viability under fluctuating demand in post-COVID-19 era: an agent-based simulation	2024	International Journal of Production Research (IJPR)	Demand forecast Production capacity and utilization (max/min capacity, capacity utilization). Order allocation and outsourcing across firms Cooperation behaviors: cooperation establishment, win-win cooperation, cooperation priority (hubs prefer other hubs). Demand fluctuation dynamics (multi-period epidemic waves)	Python	ABS	Analysis	Supply chain viability (SCV) under fluctuating demand caused by recurrent pandemic waves. Investigates how adaptive horizontal cooperation among heterogeneous manufacturers (hub LEs and non-hub SMEs) mitigates demand volatility, prevents hub failures, and improves system-level survivability.	Parameters (input) N = number of manufacturer nodes (default 1,000). Averaged degree 'K' (connectivity density of BA network). Hub selection: top 20% highest-degree nodes designated as hub manufacturers; remaining 80% are non-hub. Initial load computed from node degree and neighbor degrees; Maximum capacity, Minimum capacity, Agent order load, extra order, actual production Cooperation parameters Demand (fluctuating demand profile derived from COVID-19 waves) Performance metrics Capacity Utilization Rate Productivity Saturation On-Time Delivery Rate Relative Productivity Efficiency: ratio of total hub production capacity to non-hub production capacity Relative Order Efficiency: ratio of orders received by hubs to orders received by non-hubs. Viability indicator: proportion of surviving hub manufacturers (1 ? failed hubs / total hubs)
127	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Evaluating the supply chain resilience strategies using discrete event simulation and hybrid multi-criteria decision-making (case study: natural stone industry)	2024	Journal of Simulation (JoS)	simulation of different strategies to choose one	Arena (MATLAB used for the Bayesian BWM code)	DES	Simulation combined with hybrid MCDM: Bayesian best-worst method (BWM) for weighting the evaluation criteria and VIKOR for ranking the resilience strategies	Natural stone industry (Iran). Evaluates five supply chain resilience strategies - redundancy, backup supplier, reserved capacity, reserved inventory and increasing work-shifts. Six DES models supply the quantitative criteria, expert judgement supplies the qualitative criteria, Bayesian BWM weights the criteria and VIKOR ranks the strategies. Increasing work-shifts ranks first and reserved capacity ranks last.	five resilience strategies (redundancy, backup supplier, reserved capacity, reserved inventory, increasing work-shifts) modelled as six DES models; quantitative criteria from simulation plus qualitative criteria from experts; base model validated against company data (MAPE 5-8%) performance: total cost (\$1,559,000 in the base model), total backorders (144), fill rate (58%), recovery time (48 days), on-time deliveries (200), impact time (89 days), average response time (78 h)
128	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	Financial ripple effect in complex adaptive supply networks: an agent-based model	2024	International Journal of Production Research (IJPR)	Analysis: propagation of financial stress (liquidity squeeze from late payments) through supply networks and the structural conditions that make them vulnerable	Python	ABS	Simulation-based analysis (survivability analysis)	Agent-based supply network model with material flow, cash-constrained transactions, bargaining power and supply chain financing. Firms with higher bargaining power are more exposed to network risk; in diamond-shaped networks lower-tier firms become critical nexus suppliers.	network topology, tiers, bargaining power, repayment terms, financing cost, operation fees, market demand mean and standard deviation; performance: firm survivability

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129	Resilience Modeling, Risk Estimation and Mitigation; Supply Chain Design	Other/Not mentioned	Resilience of interdependent supply chain networks design and protection under the ripple effect	2024	International Journal of Production Research (IJPR)	Analysis: resilience of interdependent SC networks under mixed (functional and structural) ripple effects, with a network design model and a node protection strategy	Python (3.8.3)	Monte Carlo	Simulation-based analysis	Proposes a growth-maturity-decline model for designing interdependent networks and a protection strategy based on node collaboration and redundancy, evaluated on synthetic and real SC networks with 100 simulation runs. Method note: no DES, ABS or SD. The model is a complex-network cascading-failure simulation: nodes removed (randomly or by degree), their load (standing for orders) redistributed to neighbours by link weight, overloaded nodes failing with a load-based probability, iterated in steps until the cascade stops; results averaged over 100 runs. The steps are cascade rounds, not clock time: all three resilience metrics (largest connected component, weighted path length, network efficiency) are structural and taken after the cascade settles, so in effect it is a static comparison of the network before and after disruption. Unlike #72, which tracks inoperability and recovery over 90 days.	node load and capacity limit, network topology, interdependency links, failure modes, protection strategy; performance: three-dimensional resilience measure
130	Resilience Modeling, Risk Estimation and Mitigation; Inventory Analysis and Optimization	Industrial and Manufacturing	A GenAI-enhanced Framework for Agent-based Supply Chain Simulation: Integrating Unstructured Data with Strategic Decision Modelling	2025	Winter Simulation Conference (WSC)	Analysis: strategic stockpiling decisions (acquisition timing, sourcing, inventory, release triggers and quantities) for a critical mineral under market and geopolitical scenarios	Python (Mesa)	ABS	Simulation-based scenario analysis; retrieval-augmented generation extracts model inputs from unstructured documents	Six agent types from mining to reserve management. Reserves raise service level from 64.61% to 74.91% during a simulated disruption but can increase later supply variability once the buffer is exhausted.	mining, processing, trading, port, manufacturing and reserve agents, stockpile size, release rules, export bans, market state; performance: service level, stock depletion, supply variability
131	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	A supply chain digital twin for resilience analysis under immediate and long-term disruptions: a case study in plastic parts' manufacturing	2025	International Journal of Production Research (IJPR)	Analysis: resilience of a multi-echelon plastic parts SC under simultaneous long-term (economic crisis, pandemic) and short-term disruptions, and choice of recovery strategy	AnyLogistix	DES	Simulation-based scenario analysis and simulation optimisation	Generic SC digital twin framework applied to a plastic parts manufacturer over a one-year horizon. Proposes a resilience index combining profit, cost, service level and products delivered; backup inventory plus extra capacity is needed when long-term crises overlap.	suppliers, production areas, customers, monthly demand scenarios, backup inventory, production capacity, disruption type and duration; performance: profit, cost, service level, products delivered, backlog, time to recover
132	Resilience Modeling, Risk Estimation and Mitigation; Inventory Analysis and Optimization	Agriculture and Food	Assessing resilience measures in the Austrian dairy supply chain: a case-based simulation integrating strategic, tactical, and operational decisions	2025	International Journal of Production Research (IJPR)	Resilience assessment (Backup stocks. Backup supply (import options). Backup transportation. Changed reorder policies.)	AnyLogic	DES	Evaluating resilience measures across strategic, tactical, and operational levels. Addressing demand surges (panic buying) and transportation shortages (truck/fuel/driver constraints).	this article demonstrates how integrated resilience measures (backup stocks, supply, transport, policy changes, operational adaptation, early-warning) can be quantitatively assessed in a discrete-event simulation of the Austrian dairy supply chain, highlighting their effectiveness in mitigating short-term disruptions.	Parameters (inputs): Number of dairy factories (25). Number of distribution centers (31). Number of retail stores (3,868). Trucks per dairy (20) and total trucks (1,301). Processing times (24 hours at factories, 12 hours throughput). Reorder policy (order-up-to with safety stock). Routing algorithm (nearest-neighbor). Consumer demand distribution (empirical data). Disruption scenarios: Demand increase (50–100%). Transportation capacity reduction (one-third fewer trucks). Performance Metrics (outputs): Share of demand met (resilience indicator). Unmet demand percentage (during disrupted vs non-disrupted periods). Inventory positions at factories, distribution centers, and retail stores. Number of days with demand fulfillment >90%, 95%, 99%. Mean inventory levels and standard deviation. Sensitivity analysis of disruption magnitude.
133	Resilience Modeling, Risk Estimation and Mitigation; Planning, production planning	Industrial and Manufacturing	From Disruption to Profitability a Simulation Study on High Runner Focused Flexibility in Enhancing Resilience in Dual Fab Systems	2025	Winter Simulation Conference (WSC)	Analysis: which product segments (high-runners or low-runners) should be made producible at both fabs	AnyLogic	DES	Simulation-based scenario comparison	Four flexibility scenarios over 260 weeks with a 20-week processing time. Flexibility for high-runners only gives about 7% higher cumulative profit than the baseline; the gap widens at 80% forecast accuracy.	two fabs with capacity limits, six products, demand volumes, lead time, order waiting and scrapping limits, forecast accuracy; performance: cumulative profit, penalties, queue length

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134	Resilience Modeling, Risk Estimation and Mitigation; Inventory Analysis and Optimization	Other/Not mentioned	Integrating digital twin and blockchain for responsive working capital management in supply chains facing financial disruptions	2025	International Journal of Production Research (IJPR)	Analysis and optimization: working capital policies (inventory and WIP adjustment, cash collection) under trade credit extension and down payment reduction	SimPy	DES	Machine learning decision rules on simulation output; ANOVA	Integrated blockchain and digital twin framework. Both disruptions create a cash flow bullwhip that intensifies upstream; inventory adjustments are the main lever for reducing working capital variability.	customer demand, production capacity, production and distribution lead times, desired inventory and WIP, proportional controllers, cash collection policy, trade credit period, down payment; performance: cash flow bullwhip, cash conversion cycle, inventory turnover, total cost
135	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	Investigating disruption propagation and resilience of supply chain networks: interplay of tiers and connections	2025	International Journal of Production Research (IJPR)	Analysis: effect of number of tiers, intra- and inter-tier connections, redundancy and supplier diversification on disruption duration and recovery	Not specified	Other	Simulation-based analysis	A network generator creates SCNs of two to seven tiers; facility and connection disruptions are simulated. Disruptions last longer in higher tiers, redundancy cuts recovery time by up to 57% and a second supply region by up to 50%. Biofuel and solid-waste cases are in the supplement. Method note: dynamic, not static. A clock advances in steps until every disrupted node and link recovers; each step disruptions spread with probability Alpha and disrupted elements recover with probability Omega, both updated over time (Beta, Rho). Metrics come from the performance-over-time curve (resilience triangle). Time is abstract iterations; performance is the share of working network elements, with no material flow or inventory. The number of runs and the tool are not reported. Recorded as Other (closer to #72 than #26).	tiers, facilities, connections, redundancy, supplier regions, disruption of facilities and links, government support; performance: average functionality, recovery duration
136	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Modeling Supply Chain Resiliency for Assembly Fabrication Using Discrete Event Simulation	2025	Winter Simulation Conference (WSC)	Analysis: limiting components, bottleneck steps and sensitivity to contracting delays and equipment downtime in an assembly's SC	Process Simulator	DES	Simulation-based what-if analysis	Component chains are modelled in parallel with shared raw materials and process steps, and disruptions are injected at chosen times. Methodology description; no numerical results.	components, machining steps, batch sizes and times, procurement quantities and delivery frequencies, scrap rates, disruptions; performance: throughput against demand, inventory levels
137	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Optimisation of recovery policies in the era of supply chain disruptions: a system dynamics and reinforcement learning approach	2025	International Journal of Production Research (IJPR)	Optimization: adaptive order policies for recovery from disruptions and mitigation of the ripple effect	Vensim; Python (PySD)	SD	Reinforcement learning on the SD simulation; Vensim's Powell method as benchmark	An SD model derived from the literature simulates disruptions using secondary data from an aerospace and defence SC; the RL agent learns recovery order policies without full information on the disruption. Three experiments validate the approach.	SC echelons, inventory, orders, disruption location and duration, order policy; performance: reward based on service level and backlog, reliability
138	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	The multiple effects of financial squeeze: A complex adaptive system modelling with empirical data	2025	International Journal of Production Research (IJPR)	Analysis: propagation of financial squeeze and the logistics risk it triggers through an integrated supply and cash-flow network	AnyLogic	ABS	Simulation-based analysis	A transaction reconstruction algorithm builds the network from buyer-supplier and financial statement data; four company types are identified. Financial squeeze shows ripple, correlation, duality and threshold effects.	firms, buyer-supplier links, order, delivery and payment deadlines, cash, supplier and customer concentration, squeeze intensity; performance: firm and system level default and delay measures
139	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	The risk management of customized product supply chains based on digital twin technology	2025	International Journal of Production Research (IJPR)	Analysis: prediction of delivery disruption risk in a customised-product (order-driven, zero-inventory) SC and adjustment of supply plans	Plant Simulation	DES	Simulation-based prediction and plan adjustment	Suppliers are modelled at coarse, medium or fine granularity according to their effect on delivery time, with virtual-real synchronisation. Average prediction accuracy of 87.56% in enterprise projects.	suppliers, production processes, equipment efficiency, quality inspection, logistics times, order due dates; performance: predicted delivery time, prediction accuracy
140	Resilience Modeling, Risk Estimation and Mitigation; Inventory Analysis and Optimization	Healthcare and Medical	Validating Simulated Agents With Pharmaceutical Supply Chain Game Data	2025	Winter Simulation Conference (WSC)	Analysis: effect of behavioural archetypes (hoarders, reactors, followers) on ordering under disruption and information-sharing conditions; validation of simulated agents	Python	ABS	PCA, sequence clustering and hidden Markov models to derive agent behaviour; Kolmogorov-Smirnov validation	50 replications per condition are compared with gameplay data. Simulated agents match the data in most conditions; hoarders under disruption-duration information do not.	manufacturers, wholesalers, health centres, weekly orders and shipments, inventory, disruption, information shared (none, supplier capacity, duration); performance: order quantities against empirical behaviour
141	Resilience Modeling, Risk Estimation and Mitigation	Other/Not mentioned	When is the supply chain resilient? Customer and operational perspectives	2025	International Journal of Production Research (IJPR)	Analysis: on-time delivery, fill rate and time-to-recover as resilience indicators from customer and operational perspectives	AnyLogistix	DES	Simulation-based analysis	A fragment of an FMCG SC is simulated for 365 days with a 100-day warm-up and two factory disruptions. Fill rate signals disruptions immediately and suits the operational view; on-time delivery suits the customer view but reports late.	factory, distribution centres, customers, inventory and ordering policy, demand, disruption timing and duration; performance: on-time delivery, fill rate, time to recover
142	Resilience Modeling, Risk Estimation and Mitigation; Inventory Analysis and Optimization	Healthcare and Medical	A hybrid simulation methodology for identifying and mitigating supply chain disruptions	2026	Journal of Simulation (JoS)	Analysis and optimization: identifying worst-case disruptions by malicious actors and testing mitigations such as safety stock and online anomaly detection	MASON (Java)	Hybrid DES + ABS	Evolutionary optimisation (CMA-ES through ECJ) over disruptions and over safety stock	The model covers sourcing, outsourcing, production, packaging and distribution. Optimised safety stock reduces disruption impact by a factor of five; anomaly detection reaches a recall of 0.966.	suppliers, production and packaging stages, distribution, safety stock, disruption type, timing and scope, warm-up period; performance: delivered product, resilience triangle measure

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143	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	A symbiotic generative adversarial network framework for supply chain stress-testing	2026	International Journal of Production Research (IJPR)	Analysis: stress-testing a supply network with GAN-generated disruption scenarios and simulated cascading production loss	Python	Other	Conditional generative adversarial network for scenario generation; simulation-based impact assessment	Generated scenarios give 60.5% higher simulated cascading-failure impact and 64.3% higher economic loss than the training scenarios. The authors note the circularity of training and evaluating on the same simulator. Code is public. Tool note: the paper names Python only (data availability statement) and no library; the continuous-time AIM is discretised with daily time steps ($\Delta t = 1$ day, stable for $\Delta t \leq 2$). Code: https://github.com/taisai-khaki/c-GAN/tree/main .	network topology (tiers, 127 nodes), node loads and capacities, disruption vectors, persistence and frequency parameters; performance: cascading failure index, economic loss, resilience measures
144	Resilience Modeling, Risk Estimation and Mitigation; Supply Chain Design	Industrial and Manufacturing	Building resilient closed-loop supply chains: resilience enablers and diversified sourcing under global disruptions	2026	International Journal of Production Research (IJPR)	Analysis and optimization: resilience of closed-loop SC structures under local and global disruptions; proactive planning, reactive response, agility and supplier diversification	Not specified	Other	Simulation-based analysis; bi-objective optimisation of supplier diversification under a budget (GAMS)	Disruptions ripple forward, laterally and backward through the network. Diversification helps under localised disruptions but less under global ones. A biofuel closed-loop case in Victoria, Australia validates the model. The paper names no tool for the simulation itself.	facilities by tier, forward and reverse links, impact probability, recovery rate, agility parameters, number of supply regions, budget; performance: functionality index, recovery time
145	Resilience Modeling, Risk Estimation and Mitigation; Energy and Carbon Cost Modeling and Optimization	Industrial and Manufacturing	Fostering automotive Additive Manufacturing : a simulation driven decision-making approach to optimize supply chains	2026	Journal of Simulation (JoS)	Analysis: sustainability and resilience of integrating additive manufacturing at the OEM against traditional multi-tier supply	AnyLogic	ABS	Simulation-based scenario analysis	Five value streams are simulated with logistical and manufacturing disruptions. Additive manufacturing narrows the total-cost gap once end-to-end SC cost is counted and avoids import-related losses and emissions.	tier-1 and tier-2 suppliers, OEM, container shipments, transport modes, disruption timing; performance: total cost of ownership, transport carbon footprint, order fulfilment rate
146	Resilience Modeling, Risk Estimation and Mitigation; Supply Chain Design	Other/Not mentioned	Supply chain stress testing and reconfiguration for tariff shocks: connecting operational and economics perspectives	2026	International Journal of Production Research (IJPR)	Analysis and optimization: stress testing and network reconfiguration under tariff shocks, including indirect cost effects and time-to-adapt	AnyLogistix	DES	Hybrid: network optimisation (MILP) converted to DES with (s,S) inventory and LTL transport policies	Direct tariff propagation is analysed first, then indirect effects such as higher transport and production costs at nodes and routes that take peak loads after reconfiguration; results are linked to macroeconomic indicators.	suppliers, factories, distribution centres, customers, tariffs, (s,S) inventory policy, LTL shipments, capacities, time-to-adapt; performance: cost, profit, service level, lead time
147	Resilience Modeling, Risk Estimation and Mitigation	Industrial and Manufacturing	Unveiling the dynamics of organizational resilience: a systems dynamics approach	2026	Journal of Simulation (JoS)	Analysis: how sensing, seizing and reconfiguring capabilities affect preparedness, response and recovery of the SC	Vensim	SD	Simulation-based scenario analysis	Causal loop and stock-flow diagrams simulated over 2020 to 2023 with expert and behavioural validation. Visibility and collaboration build financial preparedness; agility, material flow restoration and lead time reduction improve response.	net cash flow, material inventory, capacity gap, lead time, visibility, knowledge acquisition; performance: service level, resilience by stage
148	Supply Chain Design	Agriculture and Food	An Agent-based Modelling Framework for Urban Agriculture	2019	Winter Simulation Conference (WSC)	Location (design) of supply chain nodes in order to maximize several performance metrics of the chain e.g. predict/optimize location of retailers to maximize customer satisfaction (minimize trip dist and trip time)	Repast	ABS	Study showcases simulation with different scenarios. (GIS data of agricultural supply chain is used in simulation)	The project is available in GitHub	No
149	Supply Chain Design	Humanitarian and Emergency	A Simulation Model for Short and Long Term Humanitarian Supply Chain Operations Management	2020	Winter Simulation Conference (WSC)	Design of the supply chain	Arena	DES	Simulation-based (authors model and simulate multiple designs considering different aspects of SC to find the optimum one)	Traditionally, humanitarian supply chains are designed separately for different disaster phases, (speed for immediate response and cost reduction for reconstruction) Authors propose a sustainable supply chain network integrating lean and agile principles to ensure a smooth transition from response to reconstruction.	Not explicitly mentioned
150	Supply Chain Design; Logistics Optimization and Route Planning	Other/Not mentioned	A Simulation-Optimization Approach for Locating Automated Parcel Lockers in Urban Logistics Operations	2020	Winter Simulation Conference (WSC)	To find minimum-cost number of APL to be installed every year, and estimate their locations	Not specified	SD	Simulation-based	APL: locker boxes, for which the owner and delivery person has key/code (kind of no contact delivery). A delivery person can leave the parcel in the box and lock it, and owner can get the parcel whenever he/she is available. Cost of this method is lower than home deliveries, and there are no missed deliveries.	Yes. Table summarizing parameters is presented in the paper. population, service level, number of deliveries, online purchase rate,
151	Supply Chain Design	Healthcare and Medical	Combining Simulation with Reliability Analysis in Supply Chain Project Management under Uncertainty: A Case Study in Healthcare	2021	Winter Simulation Conference (WSC)	Analysis of reliability of SC (emphasis on which nodes are connected to which, or how certain paths in SC can bottleneck) - instead of locations	Not specified	Monte Carlo	Simulation-based	A supply chain can be logically represented as multiple processing paths. Under stochastic conditions, it is hard to assess the reliability. Authors propose to build a logical representation of SC and then analyze it using simulation to assess reliability and to identify bottleneck paths	Yes. n tasks, tasks processing time t (random)
152	Supply Chain Design	Agriculture and Food; Humanitarian and Emergency	Relief Food Supply Network Simulation	2021	Winter Simulation Conference (WSC)	Analysis: to support management decision	Not specified	Other	Simulation-based analysis	a simulation model of the relief food distribution to refugees in a region that is vulnerable to multiple disasters on a daily basis. Authors model and simulate the supply chain and assess when the SC faces increased demand	disaster_type, disaster_location, n_refugee (arrival), relief_food_demand,

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153	Supply Chain Design; Logistics Optimization and Route Planning	Agriculture and Food	Simulation-based Optimization for Large-scale Perishable Agri-food Cold Chain in Rwanda: Agent-based Modeling Approach	2024	Winter Simulation Conference (WSC)	Optimization: number and location of cold storage hubs and cold chain configuration to cut food loss and emissions and raise exports	AnyLogic	ABS	Simulation-based optimization with a weighted clustering heuristic for hub locations	Thousands of scenarios compare cold-chain and non-cold-chain operation, tracking crop quality over time, waste, CO2 and export potential. Directly relevant to perishability at national scale.	farms, cold storage warehouses, vehicles, crop types and quality decay, temperature control, demand and export markets; performance: food loss, CO2 emissions, export level, cost
154	Supply Chain Design; Logistics Optimization and Route Planning	Agriculture and Food	An Agent-Based Framework for Sustainable Perishable Food Supply Chains	2025	Winter Simulation Conference (WSC)	Analysis: redesign of a centralised perishable food SC into a decentralised closed-loop network with collection points, regional hubs and biogas units	AnyLogic	ABS	Simulation-based scenario comparison	Current and proposed configurations are simulated with 10 replications. Annual transport distance and cost fall by 31.1% and cost per ton by 35.9%, with more of the transport burden shifted to retailers. Directly relevant to perishable food SCs.	farms, central market, collection points, hubs, retailers, biogas units, trucks, waste return by backhaul; performance: transport distance, transport cost, cost per ton
155	Supply Chain Design; Logistics Optimization and Route Planning	Industrial and Manufacturing	Strategic and Tactical Selection of Hubs for Rail-Based Metal Scrap Collection	2025	Winter Simulation Conference (WSC)	Analysis: number and location of scrap bundling hubs among existing shunting yards, and robustness to hub outages	AnyLogic	ABS	Simulation-based parameter variation over hub configurations and demand scenarios	One-month runs with 20 seeds, validated with the rail freight company. Fewer hubs cut transport time by up to 33%, cost by up to 9% and wagons by up to 29%, but a single hub failure in a four-hub system raises operating cost by up to 37%.	scrap senders, steel mills, hubs (shunting yards), trains and wagons, transport demand, hub outages and capacity limits; performance: transported volume, transport time, operating cost, wagons required
156	Warehouse Operations Optimization	Other/Not mentioned	A Simulation Model to Determine Staffing Strategy and Warehouse Capacity for a Local Distribution Center	2019	Winter Simulation Conference (WSC)	warehouse operation optimization to maximize warehouse capacity utilization (1) loading preparation (2) warehouse staffing (staff that load/unload boxes, sorts, puts at right places)	Simio	DES	Different scenarios are simulated and analyzed to find the optimum	Optimal Team Formation and Job Assignment to Optimize Warehouse Operations	Yes. Arrival modeled using Poisson process. Rest: specific parameters in warehouse operation. (stock Levels) e.g. Contract Type (fixed-term, perm, casual), Shift (day, night), level (expert, med, beginner)
157	Warehouse Operations Optimization; Logistics Optimization and Route Planning; Energy and Carbon Cost Modeling and Optimization	Other/Not mentioned	A simulation-based evaluation of warehouse check-in strategies for improving inbound logistics operations	2019	Simulation Modelling Practice and Theory (SIMPAT)	Warehouse operations Inbound logistics	Simio	DES	Analysis	evaluates different strategies to improve the check-in process for inbound logistics at a warehouse, aiming to reduce detention fees and CO2 emissions. The study compares current operations with proposed policies like staging areas, dynamic dispatching rules, and automated technology. The results suggest that a staging area with dynamic dispatching rules offers a cost-effective short-term solution, while automated technology provides the best long-term benefits by eliminating detention fees and significantly reducing CO2 emissions	Truck arrival rate, check-in process time, truck type and size, performance: detention fees, CO2 emission, truck turnaround time
158	Warehouse Operations Optimization; Energy and Carbon Cost Modeling and Optimization	Other/Not mentioned	A simulation-based experimental design for SBS/RS warehouse design by considering energy related performance metrics	2020	Simulation Modelling Practice and Theory (SIMPAT)	Warehouse operations	Arena	DES	Simulation-based analysis/optimization	The paper discusses a simulation-based experimental design for optimizing Shuttle-Based Storage and Retrieval Systems (SBS/RS) in warehouses, focusing on energy-related performance metrics	arrival (Poisson), speed and acceleration of lifts and shuttles, number of aisles performance: avg cycle time, energy consumption
159	Warehouse Operations Optimization	Other/Not mentioned	A simulation study on the robotic mobile fulfillment system in high-density storage warehouses	2021	Simulation Modelling Practice and Theory (SIMPAT)	Analysis	Not specified	Other	Simulation-based	investigates the Robotic Mobile Fulfillment System (RMFS) in high-density storage warehouses, focusing on improving space utilization and picking efficiency. It introduces a simulation platform to compare traditional and high-density RMFS layouts, revealing that high-density layouts can save around 10% of storage space while maintaining similar robot deployment and energy consumption levels. The study also addresses challenges like task assignment, path planning, and traffic control in high-density environments, proposing solutions for blocking and deadlock issues.	components in the model: robot, pod, zone, task (and associated param) Throughput, wait time, total travel dist, num of deadlocks
160	Warehouse Operations Optimization	Other/Not mentioned	Optimal Team Formation and Job Assignment to Optimize Warehouse Operations	2022	Winter Simulation Conference (WSC)	Optimization of warehouse operations at manufacturer (determining optimum team formation, buffer allocation and job assignments to minimize total time required to unload, scan, breakdown and store parts (which is going to be used in manufacturing)	Arena	DES	OptQuest	Authors model and simulate operations at a warehouse (unloading). The work focuses on determining the optimal team formation, buffer allocation, and job assignment to minimize the total time required to unload, scan, breakdown, and store parts from sea-containers and trailers	Number of teams, num of forklifts, workforce (teams available for various operations - scanner, forklift operator etc)

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161	Warehouse Operations Optimization	Other/Not mentioned	Assignment-simulation model for forklifts in a distribution center with aisle constraints	2024	Simulation Modelling Practice and Theory (SIMPAT)	Analysis: number and productivity of counterbalanced forklifts in a consumer goods distribution centre	Not specified	DES	Simulation-based scenario analysis	One year of warehouse management system records feeds the model, which is validated against pallets dispatched per hour. Scenarios vary forklift numbers to inform operating cost and service level.	forklift types and speeds, loading and unloading times, pallet traffic, aisle restrictions, schedules; performance: pallets moved per hour, utilisation, operating cost
162	Warehouse Operations Optimization	Other/Not mentioned	Cloud based Simulation Platform (CSP): A Novel Way to Democratize Simulation based Experimentation	2024	Winter Simulation Conference (WSC)	Optimization: staffing plans for inbound operations at distribution centres according to the inbound freight mix	FlexSim	DES	Simulation-based staffing tool with trailer selection optimisation	Describes a cloud simulation platform on Amazon Web Services (data import and export modules, simulation integration module) that lets non-specialists run DES experiments.	trailers and freight mix, conveyors and other handling equipment, operators; performance: throughput, operator utilisation, staffing requirement
163	Warehouse Operations Optimization	Other/Not mentioned	Cost and performance comparison of tier-captive SBS/RS with a novel AVS/RS/ML	2024	International Journal of Production Research (IJPR)	Analysis: throughput, lift utilisation and investment cost of a novel AVS/RS with movable lifts against a tier-captive SBS/RS	Arena	DES	Simulation-based analysis	Proposes an autonomous vehicle-based storage and retrieval system with movable lifts and compares it by simulation with a tier-captive shuttle-based system using data from an industrial mini-load warehouse. 18 design scenarios, five replications each; collision and deadlock prevention rules are modelled.	number of tiers, AGVs, movable lifts, arrival rates, rack design, collision rules; performance: throughput rate, lift utilisation, unit cost per month
164	Warehouse Operations Optimization	Other/Not mentioned	Digital Twins for Picking Processes - Cases Developed by a Brazilian Consulting Company	2024	Winter Simulation Conference (WSC)	Analysis: daily resource planning of order picking in distribution centres (DHL, Luxottica, ZF)	Simul8	DES	Simulation-based what-if analysis	Each model is fed with the day's picking load to check whether available resources can process it; the DHL model reports 98% accuracy. Case descriptions only, with no	picking lists, part locations, daily demand, hourly resources; performance: workload completed, bottlenecks, idleness
165	Warehouse Operations Optimization	Other/Not mentioned	Dominant factors in the simulation-based development of warehousing services	2024	Journal of Simulation (JoS)	Analysis: which factors dominate the performance of order picking systems when planning warehousing services in contract logistics	Not specified	DES	Simulation-based analysis with design of experiments	More than 40 factors are taken from the literature; 512 experiments with three replications per case. Demand-related factors dominate, followed by equipment specifications.	demand structure, order profile, equipment specifications, layout, operational strategies; performance: order picking system performance measures
166	Warehouse Operations Optimization	Other/Not mentioned	Metamodel-Based Order Picking for Automated Storage and Retrieval Systems	2024	Winter Simulation Conference (WSC)	Prediction: makespan of a set of picking orders in an automated storage and retrieval system	AnyLogic; Python (PyTorch)	DES	Neural network metamodel (LSTM and linear layers) trained on simulation output	3,500 independent order sets are simulated to train and test the metamodel, which predicts makespan accurately and much faster than the simulation.	order sets, SKU allocation, unit load selection, task sequencing, stochastic handling times; performance: makespan, prediction error
167	Warehouse Operations Optimization	Other/Not mentioned	Multi-method Modeling and Simulation of A Vertical Lift Module with an Integrated Buffer System Using Anylogic	2024	Winter Simulation Conference (WSC)	Analysis: throughput and cycle time of a vertical lift module integrated with a shuttle-based storage and retrieval buffer, against a stand-alone module	AnyLogic	Hybrid DES + ABS	Simulation-based comparison	Models system capacity, travel distances and velocity profiles under several conditions to find drawbacks of the integration.	trays, shuttles, lift, storage capacity, velocity profile, order arrivals; performance: throughput, cycle time
168	Warehouse Operations Optimization; Inventory Analysis and Optimization	Other/Not mentioned	Picking System Digital Twin: A Lab-based Case Study	2024	Winter Simulation Conference (WSC)	Analysis: picking policies, reorder points and worker allocation to zones in a pick-to-light system	Simio	DES	No optimization; a framework for student experiments	The model mirrors the physical lab system in near real time from live and historical data; a single-picker scenario is run in parallel with the physical system.	picking zones, workers, orders, inventory and reorder point; performance: picking time, system status indicators
169	Warehouse Operations Optimization	Other/Not mentioned	Reducing Transient Behavior in Simulation-Based Digital Twins: A Novel Initialization Approach for Order Picking Systems	2024	Winter Simulation Conference (WSC)	Modelling: initialising a warehouse simulation with the real system's load state to shorten transient behaviour	Plant Simulation	DES	No optimization; comparison of initialised and empty-start runs	The initialisation concept loads released orders and resources from SAP Extended Warehouse Management and clearly shortens the transient phase compared with an empty start.	picking areas, released orders, resources, load state at start; performance: length of transient phase, throughput measures
170	Warehouse Operations Optimization	Other/Not mentioned	Simulation and process mining in a cross-docking system: a case study	2024	International Journal of Production Research (IJPR)	Optimization: number of doors and loaders in a cross-docking centre of a chain store	Arena	DES	Simulation-based scenario analysis; process mining (heuristic and inductive miner, PM4PY) for process discovery	RFID event logs are mined to discover the real process, conformance is checked, and the organisational, discovered and improved processes are simulated in Arena to find the best configuration of doors and loaders.	inbound and outbound doors, loaders, product types, process times; performance: resource utilisation, waiting and throughput times
171	Warehouse Operations Optimization	Other/Not mentioned	A Federated Simulation Framework for Modeling Complex Logistics: Amazon Sort Center Case Study	2025	Winter Simulation Conference (WSC)	Modelling: decomposing a sort centre simulation into inbound, main sortation and outbound, non-conveyable, and non-compliant federates	FlexSim	DES	No optimization; a distributed simulation framework	Federates synchronise through a conservative time-stepping algorithm and message passing. Reports 40% less model development time and 15% faster execution. Case description with little experimental detail.	trailers, packages, conveyors and sorters, mobile robots, manual processes; performance: throughput, bottlenecks, development and run time
172	Warehouse Operations Optimization	Other/Not mentioned	Agent-based modelling for human–robot collaborative order picking system considering workers' performance	2025	International Journal of Production Research (IJPR)	Analysis: effect of human roles, fatigue and learning on a collaborative human-AGV order picking system	Python	ABS	Simulation-based analysis; mathematical model (DOcplex) for AGV path planning and a matching	Three scenarios of human roles are simulated for 8 hours after a 1-hour warm-up with varying numbers of pickers and AGVs. The best scenario depends on the ratio of humans to AGVs, and safety risk rises with AGV	warehouse layout, number of pickers and AGVs, orders, fatigue and learning curve; performance: cycle time, AGV utilisation, safety indicator
173	Warehouse Operations Optimization	Industrial and Manufacturing	Development of a Library of Modular Components to Accelerate Material Flow Simulation in the Aviation Industry	2025	Winter Simulation Conference (WSC)	Modelling: reusable process blocks and resource agents that speed up building logistics simulation models	AnyLogic	Hybrid DES + ABS	No optimization; model library with a validation case	Resources are agents and processes are flowchart blocks, each validated separately. The warehouse model is checked against delivery, stock and picking data.	layout, resources, load carriers, deliveries, initial stock, picking parameters, shift model; performance: outbound and picking performance

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174	Warehouse Operations Optimization	Other/Not mentioned	Integrating storage allocation with manual order picking and replenishment operations in a distribution centre	2025	International Journal of Production Research (IJPR)	Optimization: storage location assignment integrated with manual order picking and replenishment in a wholesale distribution centre	Custom code in Microsoft Visual Studio 17.4 (language not named); CPLEX 20.1.0 benchmark	DES	Simulation-based optimization: iterated local search with embedded DES; CPLEX as benchmark	The DES reproduces several order pickers under uncertainty, interference and congestion with an S-shape policy. With tuned parameters the framework improves warehouse productivity by about 17% on real data. Tool note: the DES is custom-built from an event graph (Schruben 1983) and run in Microsoft Visual Studio 17.4.14; the programming language is not named. It has a GUI, and the warehouse map is read from an Excel file exported from the company's AS400 system. The event graph is deferred to a companion paper, likely #155.	storage and replenishment positions, item rotation, order pickers, picking policy, congestion; performance: travel distance, items picked per day, operational times
175	Warehouse Operations Optimization	Other/Not mentioned	Performance assessment of a multi-level shuttle system: an optimisation-embedded simulation study on time and energy consumption	2025	Journal of Simulation (JoS)	Analysis: travel time and energy consumption of a multi-level shuttle system across warehouse configurations, load handling capacity and optimisation settings	AnyLogic	DES	Dynamic programming for sequencing storage and retrieval missions, embedded in the simulation	The conceptual model is a UML class diagram; each configuration is simulated for an 8-hour shift with stochastic handling times. Handling two unit loads at once is advantageous in every configuration tested.	rack configuration and capacity, unit load types, number of loads handled, handling times, size of optimisation session; performance: cycle time, energy consumption, throughput
176	Warehouse Operations Optimization	Industrial and Manufacturing	Safety and Productivity Validation through Fast Iterative Discrete-Event Simulation with a Lightweight Industrial Simulator	2025	Winter Simulation Conference (WSC)	Analysis: warehouse layout, number of forklifts by type and number of workers, for both safety and lead time	Not specified	DES	Simulation-based scenario analysis	A heat map of human-forklift proximity identifies the riskiest location (countermeasure: fewer racks); a shortage of some forklift types is the bottleneck in kitting.	layout and racks, forklift types and numbers, workers, kitting tasks; performance: lead time, utilisation, collision risk
177	Warehouse Operations Optimization	Other/Not mentioned	Simulating Inventory Slotting Strategies in a Robotic Mobile Fulfillment Center	2025	Winter Simulation Conference (WSC)	Analysis: random, class-based and order-correlation-based inventory slotting strategies	Simio	DES	Simulation-based comparison	Inspired by a shoe fulfilment centre and run on synthetic data. Preliminary tests show that slotting affects cycle time, pile-on and AGV travel, more so at the larger scale.	shelf positions, pickers, stockers, AGVs, SKUs with long-tail popularity, orders of 1 to 5 SKUs; performance: order cycle time, pile-on, AGV travel distance, picker utilisation, orders within service level agreement
178	Warehouse Operations Optimization	Other/Not mentioned	Simulation of Amazon Inbound Cross Dock Facilities	2025	Winter Simulation Conference (WSC)	Analysis: cross-belt sorter recirculation settings in current buildings and design of a future automation concept	FlexSim	DES	Simulation-based sensitivity analysis	Modules for flow control, workstations and inbound and outbound processing. The model is compared with historical hourly volumes, sorter utilisation and travel times, and used to size work cells, conveyor accumulation and diverts. Case description with little numerical detail.	trailer arrivals and dock doors, staffing plan, workstation rates, sorter and conveyor parameters, recirculation limit, destination mix; performance: throughput, sorter utilisation, recirculation rate
179	Warehouse Operations Optimization	Industrial and Manufacturing	Using Discrete Event Simulation to Evaluate the Impact of Layouts and Resource Allocation in Packaging Systems	2025	Winter Simulation Conference (WSC)	Analysis: whether a connector to an adjacent warehouse and extra forklifts raise throughput of four packaging lines	FlexSim	DES	Simulation-based experiments with a metamodel	Two dock doors are the bottleneck at about 250 pallets a day; the connector adds about 30 pallets a day (8%) and makes extra forklifts worthwhile. One intersection carries 90.4 forklift trips an hour.	packaging lines, 22 material inputs, pallet storage and staging positions, dock doors, forklifts and task priorities, machine repairs; performance: pallets per day, dock door status, forklift trips
180	Warehouse Operations Optimization; Logistics Optimization and Route Planning	Other/Not mentioned	Integrated layout design and space allocation in multi-purpose terminals: a bi-level simulation optimisation approach	2026	International Journal of Production Research (IJPR)	Optimization: warehouse layout and storage space allotment to shippers in a multi-purpose maritime terminal under operational uncertainty	O2DES.Net	DES	Simulation-based optimization: particle swarm optimisation with bi-level optimal computing budget allocation	The bi-level approach improves convergence by 2.5% and 23.7% and total cost by 5.2% and 10.9% against two single-level benchmarks. Shipper behaviour (cargo load per truck, truck interarrival time) shapes the preferred layout.	warehouse designs and locations, shippers, truck interarrival time, cargo load per truck, storage demand; performance: total cost including opportunity cost, productivity
181	Warehouse Operations Optimization	Other/Not mentioned	Simulation-based performance analysis of a Robotic Compact Storage and Retrieval System under single-robot operation	2026	Simulation Modelling Practice and Theory (SIMPAT)	Analysis: effect of total bin capacity, stack height, arrival rate and robot velocity on a single-robot compact storage system	Arena	DES	Simulation-based full-factorial experiment with ANOVA	20 replications per scenario; the model is checked against earlier analytical results (error below 0.1% in most configurations). Robot velocity is the dominant factor, then bin capacity and arrival rate; stack height matters little.	grid size and bin capacity, stack height, order arrival rate, robot velocity, acceleration, bin relocation rule; performance: cycle time, robot utilisation, queue measures
182	Warehouse Operations Optimization	Other/Not mentioned	Towards a digital twin solution for manual order-picking operations in a wholesale distribution center	2026	Simulation Modelling Practice and Theory (SIMPAT)	Analysis: order-picking control policies ('skip and go' against 'stop and queue') and picker interference in parallel aisles	Not specified	DES	Simulation-based policy comparison	Models an S-shaped person-to-goods strategy for a typical workday at a large Italian retail cooperative, with 10 replications. Companion to #43 by the same group.	aisles, picking slots, pickers and vehicles, orders, inventory loaded from file, control rule; performance: picker productivity, order response time, queueing
183	Other	Other/Not mentioned	Hybrid Optimization Approach for Supply Chain Planning	2019	International Conference on Management Science and Industrial Engineering (MSIE)	Optimization (case study): maximizing profit while meeting makespan and service level requirement	Arena	Other	Hybrid: Simulation + Analytical CPLEX optimization software	Authors propose a hybrid approach simulation+analytical for supply chain planning	Demand, product costs, holding costs, transport costs, process time (production), workload capacity

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184	Other	Industrial and Manufacturing	Integrated Planning of Production and Engineering Activities in Semiconductor Supply Chains: A Simulation Study	2019	Winter Simulation Conference (WSC)	planning production and engineering activities to minimizing cost (diff production and planning costs are considered)	AutoSched	Other	MIMAC 1 model (by some other authors) is extended to incorporate engineering activities	optimizing planning for production activities and engineering activities in semiconductor supply chain	Yes. Specific attributes: production product index, ... (list of parameters specific to semiconductor manufacturing process)
185	Other	Industrial and Manufacturing	Simulation-Based Analysis of the Nervousness within Semiconductor Supply Chain Planning: Insight from a Case Study	2019	Winter Simulation Conference (WSC)	Analysis	Not specified	Hybrid DES + ABS	Analysis	Analysis of instability of demand in SC caused by lack of compatibility between planning and real production in semiconductor supply chains	No
186	Other	Other/Not mentioned	Supply chain hybrid simulation: From Big Data to distributions and approaches comparison	2019	Simulation Modelling Practice and Theory (SIMPAT)	Material information flow at a supplier (supplier orders, material arrivals and internal arrivals)	Simio	ABS	Analysis	a hybrid simulation model for supply chains, integrating Big Data and statistical distributions to manage uncertainty and variability. It highlights the benefits of using Big Data technologies to enhance simulation models, allowing for more accurate risk scenario testing and predictions. The proposed model can run using data from a Big Data Warehouse, statistical distributions, or a combination of both, providing flexibility and improved computational efficiency. The paper also compares these approaches, emphasizing their pros and cons, and sets a foundation for future research in supply chain simulation within Big Data contexts.	supplier order interarrival time, order quantity, production order interarrival time, lead time num of order placed, total quantities ordered, received and consumed
187	Other	Other/Not mentioned	Designing a stochastic multi-objective simulation-based optimization model for sales and operations planning in built-to-order environment with uncertain distant outsourcing	2020	Simulation Modelling Practice and Theory (SIMPAT)	Sales and operations planning, Inventory	Arena	DES	Simulation-based optimization	It aims to balance logistics costs and customer experience by evaluating two inventory strategies and optimizing flexibility and safety stock levels. The model uses a simulation-based approach to handle the complexity of supply chain systems, offering insights into managing demand-supply integration and responsiveness investigates the consequences of stochastic disruptions in terms of demands uncertainty, lost sales threshold, unreliable distant outsourcing, and corresponding costs.	demand, inventory parameters and policies, safety stock etc performance: logistics cost, customer experience level (table listing specific parameters is present in paper)
188	Other	Agriculture and Food	A simulation model to improve the value of rice supply chain (A case study in East Java – Indonesia)	2022	Journal of Simulation (JoS)	simulation based optimization	Not specified	SD	Simulation-based scenario analysis (policy scenarios: BioIndustry programme, farmer empowerment, information sharing)	Rice supply chain in East Java, Indonesia. System dynamics model (causal loop and stock-and-flow diagrams) to improve the value of the rice supply chain. Increasing the quantity and quality of crop yield together with the BioIndustry programme, farmer empowerment and information sharing raises rice productivity by about 1.4% per year; selling rice directly and implementing the BioIndustry programme raises total farmer profit by about 7.6% per year.	harvested area, land productivity, dry grain and rice production; crop yield quantity and quality; mileages between supply chain actors, normally distributed (agricultural area to collector, collector to rice milling unit, RMU to local retailers and to outside producer areas); farmer profit and profit from rice byproducts performance: value of the rice supply chain, rice productivity (about +1.4%/year), total farmer profit (about +7.6%/year)
189	Other	Healthcare and Medical; Humanitarian and Emergency	Calibrating Simulation Models with Sparse Data: Counterfeit Supply Chains During COVID-19	2022	Winter Simulation Conference (WSC)	Simulation model of the supply chain is calibrated to increase the similarity between the model and the system	Not specified	DES	Model calibration tunes and estimates parameters using observed data. Genetic Algorithms and Bayesian Inference are two calibration techniques.	The study evaluates how well these calibration techniques work for a counterfeit PPE simulation model. Counterfeit supply chains refer to networks where counterfeit or unauthorized goods infiltrate legitimate supply chains.	manufacture time, lead time, inventory levels, transportation distances
190	Other	Healthcare and Medical; Humanitarian and Emergency	Covid-19 Supply Chain Planning: A Simulation-optimization Approach	2022	Winter Simulation Conference (WSC)	Inventory and resources (beds, ventilators) optimization during pandemic (COVID19) (to maximize availability)	Simio	Other	Optimization (OptQuest)	The aim is to determine the optimum replenishment policy (S,s) to stock personal protective equipment (PPE) (masks, gloves etc) i.e. Optimum values of (S,s)	Contagion Factor, Beds Capacity, Ventilator Capacity, Social Distancing Factor, Reported Cases, Service Area Population:, Contagion Duration: Uniform Hospital Stay Length: Triangular

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191	Other	Agriculture and Food	Eliciting agents' behaviour using scenario-based questionnaire in agent-based dairy supply chain simulation	2022	Journal of Simulation (JoS)	Data collection and model parameterisation: eliciting agents' decision rules with a scenario-based questionnaire, and its effect on the validity of the agent-based model	Not specified	ABS	No optimization. Model validation: micro-validity of the elicited decision rules and macro-validity measured as the mean error between model output and real data (2010-2012)	Dairy supply chain in Pangalengan, West Java, Indonesia (KPBS cooperative). Designs a scenario-based questionnaire to elicit the behaviour of agents for agent-based modelling and simulation and applies it to a dairy supply chain case. Respondents respond well to the scenario-based questionnaire because it relates closely to their actual decision-making process, and the decision rules extracted this way improve the validity of the ABMS model.	agent attributes with fitted distributions: farmer age (Triangular, 22-74 years), family labour (Binomial), number of cows (Poisson), number of bulls (Poisson), peak milk production in litres (Normal); cattle population, cow population and average daily milk production for 2010-2012 performance: mean error (ME) and its standard deviation between model output and real data; micro- and macro-validity of the agent-based model
192	Other	Agriculture and Food; Humanitarian and Emergency	Modeling and Simulation of Food Bank Disaster Relief Operations	2022	Winter Simulation Conference (WSC)	Analysis of diff distribution policies in Food bank disaster supply chains to increase demand fulfillment	Simio	DES	Analysis: simulation-based method Authors model and simulate three different distribution policies	Authors discuss use of DES to model and simulate the operations of food banks at supply chain level. They simulate flow of donations and demand for supplies at 3 banks before and after a natural disaster	Resource capacities (entry/exit doors, trucks, forklifts) Demand (arrival of donations)
193	Other	Healthcare and Medical; Humanitarian and Emergency	Simulating Counterfeit Personal Protective Equipment (PPE) Supply Chains During Covid-19	2022	Winter Simulation Conference (WSC)	Analysis: Counterfeit PPE supply chain behavior is modeled with disruptions	Not specified	DES	Analysis	Law enforcements must identify the most effective disruption strategies to combat counterfeit SC, so to protect health frontline workers as well as the general population Authors propose a framework that aims to identify effective enforcement actions (disruptions) to combat counterfeit PPE supply chains, considering uncertainties	Not explicitly
194	Other	Other/Not mentioned	A Simulation-Based TDABC Model to Manage Supply Chain Costing: A Case Study	2023	Winter Simulation Conference (WSC)	No analysis or optimization	AnyLogic	DES	None	Authors build a simulation-based costing framework (to account for real time events happening in a supply chain that contributes to SC costs) that integrates DES and Time-Driven Activity-based Costing	Specific: capacity cost (department cost over total capacity supplied) truck arrival rate, unloading time, inspect time, resource availability
195	Other	Industrial and Manufacturing; Healthcare and Medical	Investigating Production Yield Effect on Inventory Control Through a Hybrid Simulation Approach	2023	Winter Simulation Conference (WSC)	Analysis: Authors model manufacturing process with uncertainty to improve production planning	AnyLogic	Hybrid DES + ABS	Analysis	This paper explores how to improve production planning and control (PPC) in manufacturing by using a hybrid simulation approach. Authors model a contact lens manufacturing process with uncertain yield rates. a simple average yield for planning can lead to backlogs and extra inventory costs, hybrid simulation helps create more accurate schedules	order, initial yield, production yield, packaging yield, inventory status
196	Other	Other/Not mentioned	Simulating the Impact of Forecast related Overbooking and Underbooking Behavior on MRP Planning and a Reorder Point System	2023	Winter Simulation Conference (WSC)	Optimization: authors assess the performance of production planning methods (MRP, RPS)	Not specified	DES	Optimization	Comparison between performance of MRP (material req planning) and RPS (reorder point systems) under demand uncertainty (underbooking and overbooking)	safety stock, lot size, planned lead time, reorder point, forecast uncertainty, forecast bias
197	Other	Agriculture and Food	Discrete-event simulation to analyse the impacts of a reduced minimum order quantity in the food industry: proposing a shared-savings contract	2024	Journal of Simulation (JoS)	Order quantity: impact of a reduced minimum order quantity (MOQ) on production performance, and design of a shared-savings contract	AnyLogic	DES	Simulation-based analysis (scenario comparison of four production plans); Big Data analysis and correlation analysis used to derive the input distributions	Food processing industry, cheese processing company. Data-driven DES evaluates the cost development resulting from a reduced minimum order quantity. Production costs increase significantly with higher breakdowns, setups and rework, and a 20% increase in orders requires about 3% additional production capacity. Proposes a shared-savings contract in which the retailer shares part of its cost savings with the manufacturer, and concludes that transparent information sharing is key.	minimum order quantity (MOQ); order volume (+20% scenario); four production plans (A-D); breakdown and rework distributions derived from Big Data (right-skewed); setup time; rework modelled as a function of down time; production capacity performance: breakdown time, setup time, rework amount, rework time, production cost
198	Other	Healthcare and Medical	A simulation-based approach for reconstructing a diverse set of supply chain models with sparse data using a quality diversity algorithm	2026	Simulation Modelling Practice and Theory (SIMPAT)	Calibration: generating a diverse set of plausible SC simulation models (structure and parameters) that fit sparse observations	Python (pydsol)	DES	Quality diversity algorithm for model calibration	A stylised counterfeit PPE SC serves as ground truth and data sparseness is varied; 10 replications per evaluation. The algorithm recovers the ground truth best at low sparseness and overfits to more complex networks as sparseness grows. Follows the counterfeit SC calibration papers already in the corpus.	SC structure (actors and links), transport times and costs, time intervals, degree of data sparseness; performance: distance between simulated and observed data, transport cost, network vulnerability